brackets

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Disquisitiones generales circa seriem infinitam

$$1+\frac{\alpha\beta}{1\cdot\gamma}x+\frac{\alpha(\alpha+1)\beta(\beta+1)}{1\cdot2\cdot\gamma(\gamma+1)}xx+\frac{\alpha(\alpha+1)(\alpha+2)\beta(\beta+1)(\beta+2)}{1\cdot2\cdot3\cdot\gamma(\gamma+1)(\gamma+2)}x^3+\text{etc.}$$

Sectio Tertia.

De summa seriei nostrae statuendo elementum quartum $= 1$, ubi simul quaedam aliae functiones transscendentes discutiuntur.

15.

Quoties elementa α, β, γ omnia sunt quantitates positivae, omnes coefficientes potestatum elementi quarti x positivi evadunt: quoties vero ex illis elementis unum alterumve negativum est, saltem inde ab aliqua potestate x^m omnes coefficientes eodem signo affecti erunt, si modo m accipitur maior quam valor absolutus elementi negativi maximi.

Porro hinc sponte patet, seriei summam pro $x = 1$ finitam esse non posse, nisi coefficientes saltem post certum terminum in infinitum decrescant, vel, ut secundum morem analystarum loquamur, nisi coefficiens termini x^m sit $= 0$. Ostendemus autem, et quidem, in gratiam eorum, qui methodis rigorosis antiquorum geometrarum favent, omni rigore,

primo, coefficientes (siquidem series non abrumpatur), in infinitum crescere, quoties fuerit $\alpha+\beta-\gamma-1$ quantitas positiva.

secundo, coefficientes versus limitem finitum continuo convergere, quoties fuerit $\alpha+\beta-\gamma-1=0$.

tertio, coefficientes in infinitum decrescere, quoties fuerit $\alpha+\beta-\gamma-1$ quantitas negativa.

quarto, summam seriei nostrae pro $x = 1$, non obstante convergentia in casu tertio, infinitam esse, quoties fuerit $\alpha+\beta-\gamma$ quantitas positiva vel $= 0$.

quinto, summam vero finitam esse, quoties $\alpha+\beta-\gamma$ fuerit quantitas negativa.

16.

Hanc disquisitionem generalius adaptabimus seriei infinitae $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc. ita formatae, ut quotientes $\frac{\mathfrak{M}'}{\mathfrak{M}}, \frac{\mathfrak{M}''}{\mathfrak{M}'}, \frac{\mathfrak{M}'''}{\mathfrak{M}''}$ etc. resp. sint valores fractionis

$$\frac{t^\lambda+At^{\lambda-1}+Bt^{\lambda-2}+Ct^{\lambda-3}+\text{etc.}}{t^\lambda+at^{\lambda-1}+bt^{\lambda-2}+ct^{\lambda-3}+\text{etc.}}$$

pro $t = m$, $t = m+1$, $t = m+2$ etc.

Brevitatis caussa huius fractionis numeratorem per P, denominatorem per p denotabimus: praeterea supponemus, P, p non esse identicas, sive differentias $A-a$, $B-b$, $C-c$ etc. non omnes simul evanescere.

I. Quoties e differentiis $A-a$, $B-b$, $C-c$ etc. prima quae non evanescit est positiva, assignari poterit limes aliquus l, quem simulac egressus est valor ipsius t, valores functionum P et p certo semper evadent positivi, atque $P>p$. Manifestum est, hoc evenire, si pro l accipiatur radix maxima realis aequationis $p(P-p)=0$; si vero haec aequatio nullas omnino radices reales habeat, proprietatem illam pro omnibus valoribus ipsius t locum habere. Quapropter in serie $\frac{\mathfrak{M}'}{\mathfrak{M}}\cdot\frac{\mathfrak{M}''}{\mathfrak{M}'}\cdot\frac{\mathfrak{M}'''}{\mathfrak{M}''}$ etc. saltem post certum intervallum (si non ab initio) omnes termini erunt positivi atque maiores unitate; quodsi itaque nullus neque $=0$ neque infinite magnus evadit, perspicuum est, seriem $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc. si non ab initio tamen post certum intervallum omnes suos terminos eodem signo affectos continuoque crescentes habituram esse.

Eadem ratione, si e differentiis $A-a$, $B-b$, $C-c$ etc. prima quae non evanescit est negativa, series $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc. si non ab initio tamen post certum intervallum omnes suos terminos eodem signo affectos continuoque decrescentes habebit.

II. Si iam coefficientes A, a sunt inaequales, termini seriei $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc. ultra omnes limites sive in infinitum vel crescent vel decrescent, prout differentia $A-a$ est positiva vel negativa: hoc ita demonstramus. Si $A-a$ est quantitas positiva, accipiatur numerus integer h ita, ut fiat $h(A-a)>1$, statuaturque $\frac{\mathfrak{M}^{(h)}}{\mathfrak{M}^h}=N$, $\frac{\mathfrak{M}'^{(h)}}{\mathfrak{M}'^h}=N'$, $\frac{\mathfrak{M}''^{(h)}}{\mathfrak{M}''^h}=N''$, $\frac{\mathfrak{M}'''^{(h)}}{\mathfrak{M}'''^h}=N'''$ etc., nec non $t^{h\lambda}P=Q$, $(t+1)^{h\lambda}p^{(h)}=q$. Tunc patet, $\frac{N'}{N}\cdot\frac{N''}{N'}\cdot\frac{N'''}{N''}$ etc. esse valores fractionis $\frac{Q}{q}$ ponendo $t=m$, $t=m+1$, $t=m+2$ etc., ipsas Q, q vero esse functiones algebraicas formae huius

$$\begin{aligned} Q &= t^{h\lambda+hA} + \text{etc.} \\ q &= t^{h\lambda+ha+1} + \text{etc.} \end{aligned}$$

Quare quum per hyp. differentia $hA-(ha+1)$ sit quantitas positiva, termini seriei N,N',N'',N''' etc. si non ab initio tamen post certum intervallum continuo crescent (per I); hinc termini seriei $mN,(m+1)N',(m+2)N'',(m+3)N'''$ etc. necessario ultra omnes limites crescent, et proin etiam termini seriei $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc., quippe quorum potestates exponente h illis sunt aequales. Q. E. P.

Si $A-a$ est quantitas negativa, accipere oportet integrum h ita, ut $h(a-A)$ fiat maior quam 1, unde per ratiocinia similia termini seriei

$$m\mathfrak{M}^h,\quad (m+1)\mathfrak{M}'^h,\quad (m+2)\mathfrak{M}''^h,\quad (m+3)\mathfrak{M}'''^h\quad \text{etc.}$$

post certum intervallum continuo decrescent. Quamobrem termini seriei $\mathfrak{M}^{(h+1)}$, $\mathfrak{M}'^{(h+1)}$, $\mathfrak{M}''^{(h+1)}$ etc., adeoque etiam termini huius $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc. necessario in infinitum decrescent. Q. E. S.

III. Si vero coefficientes primi A, a sunt aequales, termini seriei $\mathfrak{M},\mathfrak{M}',\mathfrak{M}'',\mathfrak{M}'''$ etc. versus limitem finitum continuo convergent, quod ita demonstramus. Supponamus primo, terminos seriei post certum intervallum continuo crescere, sive e differentiis $B-b$, $C-c$ etc. primam quae non evanescat esse positivam. Sit h integer talis, ut $h+hB-b$ fiat quantitas positiva, statuamusque

$$\mathfrak{M}(\tfrac{t+1}{t})^h=N,\quad \mathfrak{M}'(\tfrac{t+2}{t+1})^h=N',\quad \mathfrak{M}''(\tfrac{t+3}{t+2})^h=N''\quad \text{etc.}$$

atque $(tt-1)^\lambda P=Q$, $t^\lambda p=q$, ita ut $\frac{N'}{N}$, $\frac{N''}{N'}$ etc. sint valores fractionis $\frac{Q}{q}$ ponendo $t=m$, $t=m+1$ etc.

Quum itaque habeatur

$$\frac{Q}{q} = \frac{t+1}{t} \cdot \frac{(t-1)^{\lambda}P}{t^{\lambda}p} \cdot \frac{t^h}{(t-1)^h} = \frac{(t+1)\cdot(t-1)^{\lambda-h}P}{t^{\lambda-h+1}p}$$
$$= \frac{t^{\lambda+1} + (A-h-m+1)t^{\lambda} + \text{etc.}}{t^{\lambda+1} + (a-m)t^{\lambda} + \text{etc.}}$$

atque per hyp. $B-h-b$ sit quantitas negativa, termini seriei N, N', N'', N''' etc. post certum saltem intervallum continuo decrescent, adeoque termini seriei $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc., qui illis resp. semper sunt minores, dum continuo crescunt, tantummodo versus limitem finitum convergere possunt. Q. E. P.

Si termini seriei $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc. post certum intervallum continuo decrescunt, accipere oportet pro h integrum talem, ut $h+B-b$ sit quantitas positiva, evinceturque per ratiocinia prorsus similia, terminos seriei

$$\mathfrak{M}(\tfrac{t}{t+1})^h, \quad \mathfrak{M}'(\tfrac{t+1}{t+2})^h, \quad \mathfrak{M}''(\tfrac{t+2}{t+3})^h \quad \text{etc.}$$

post certum intervallum continuo crescere, unde termini seriei $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}''$ etc., qui illis resp. semper sunt maiores, dum continuo decrescunt, necessario tantummodo versus limitem finitum decrescere possunt. Q. E. S.

IV. Denique quod attinet ad summam seriei, cuius termini sunt $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc., in casu eo, ubi hi in infinitum decrescunt, supponamus primo, $A-a$ cadere inter 0 et -1, sive $A+1-a$ esse vel quantitatem positivam vel $= 0$. Sit h integer positivus, arbitrarius in casu eo, ubi $A+1-a$ est quantitas positiva, vel talis qui reddat quantitatem $h+m+A+B-b$ positivam in casu eo, ubi $A+1-a=0$. Tunc erit

$$P(t-(m+h-1)) = t^{\lambda+1} + (A+1-m-h)t^{\lambda} + (B-A(m+h-1))t^{\lambda-1} + \text{etc.}$$
$$p(t-(m+h)) = t^{\lambda+1} + (a-m-h)t^{\lambda} + (b-a(m+h))t^{\lambda-1} + \text{etc.}$$

ubi vel $A+1-m-h-(a-m-h)$ erit quantitas positiva, vel, si haec fit $= 0$, saltem $B-A(m+h-1)-(b-a(m+h))$ positiva erit. Hinc (per I) pro quantitate t assignari poterit valor aliquus l, quem simulac transgressa est, valores fractionis $\frac{P(t-(m+h-1))}{p(t-(m+h))}$ semper fiunt positivi atque unitate maiores. Sit n integer maior quam l simulque maior quam h, sintque termini seriei $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc. qui respondent valoribus $t=m+n$, $t=m+n+1$, $t=m+n+2$ etc., hi N, N', N'', N''' etc. Erunt itaque

$$\frac{(n+1-h)N'}{(n-h)N}, \quad \frac{(n+2-h)N''}{(n+1-h)N'}, \quad \frac{(n+3-h)N'''}{(n+2-h)N''} \quad \text{etc.}$$

quantitates positivae unitate maiores, unde

$$\frac{N'}{N} > \frac{n-h}{n+1-h}, \qquad \frac{N''}{N'} > \frac{n+1-h}{n+2-h}, \qquad \frac{N'''}{N''} > \frac{n+2-h}{n+3-h} \quad \text{etc.}$$

adeoque summa seriei $N+N'+N''+N'''+$ etc. maior summa seriei

$$(n-h)N\Big(\tfrac{1}{n-h} + \tfrac{1}{n+1-h} + \tfrac{1}{n+2-h} + \tfrac{1}{n+3-h} + \text{etc.}\Big)$$

quotcunque termini colligantur. Sed posterior series, crescente terminorum numero in infinitum, omnes limites egreditur, quum summa seriei $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+$ etc. quam infinitam esse constat etiam infinita maneat, si ab initio termini $1+\frac{1}{2}+\frac{1}{3}+$ etc. $+\frac{1}{n-h}$ rescindantur.

Quare summa seriei $N + N' + N'' + N'''+$ etc., adeoque etiam summa huius $\mathfrak{M} + \mathfrak{M}' + \mathfrak{M}'' + \mathfrak{M}'''+$ etc., cuius pars est illa, ultra omnes limites crescit.

V. Quoties autem $A - a$ est quantitas negativa absolute maior quam unitas, summa seriei $\mathfrak{M} + \mathfrak{M}' + \mathfrak{M}'' + \mathfrak{M}'''+$ etc. in infinitum continuatae certo erit finita. Sit enim h quantitas positiva minor quam $a - A - 1$, demonstrabiturque per ratiocinia similia, assignari posse valorem aliquem l quantitatis t, ultra quem fractio $\frac{P}{p}$ semper adipiscatur valores positivos unitate minores. Quodsi iam pro n accipitur numerus integer ipsis l, m, $A+1$ maior, terminiq ue seriei $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}'', \mathfrak{M}'''$ etc., valoribus $t = n$, $t = n+1$, $t = n+2$ etc. respondentes, designantur per N, N', N'' etc., erit

$$\frac{N'}{N} < \frac{n-h-1}{n}, \qquad \frac{N''}{N'} < \frac{(n-h-1)(n-h)}{n(n+1)},$$
$$\frac{N'''}{N''} < \frac{(n-h-1)(n-h)(n-h+1)}{n(n+1)(n+2)} \quad \text{etc.}$$

adeoque summa seriei $N + N' + N''+$ etc., quotcunque termini colligantur, minor producto ex N in summam totidem terminorum seriei

$$1 + \frac{n-h-1}{n} + \frac{(n-h-1)(n-h)}{n(n+1)} + \frac{(n-h-1)(n-h)(n-h+1)}{n(n+1)(n+2)} + \text{etc.}$$

Huius vero summa pro quolibet terminorum numero facile assignari potest; est scilicet

$$\begin{aligned} &\text{terminus primus} && = \tfrac{n}{n-h-1}, \\ &\text{summa duorum terminorum} && = \tfrac{n(n-h)}{(n-h-1)(n-h)}, \\ &\text{summa trium terminorum} && = \tfrac{n(n-h)(n-h+1)}{(n-h-1)(n-h)(n-h-1)}, \\ &\text{etc.} \end{aligned}$$

et quum pars altera (per II) formet seriem ultra omnes limites decrescentem, summa illa in infinitum continuata statui debet $= \frac{n}{n-h-1}$. Hinc $N + N' + N''$ etc. in infinitum continuata semper manebit minor quam $\frac{nN}{n-h-1}$ et proin $\mathfrak{M} + \mathfrak{M}' + \mathfrak{M}''$ etc. certo ad summam finitam converget. Q. E. D.

VI. Ut ea, quae generaliter de serie $\mathfrak{M}, \mathfrak{M}', \mathfrak{M}''$ etc. demonstravimus, ad coefficientes potestatum x^m, x^{m+1}, x^{m+2} etc. in serie $F(\alpha, \beta, \gamma, x)$ applicentur, statuere oportebit $\lambda = 2$, $A = \alpha + \beta$, $B = \alpha\beta$, $a = \gamma + 1$, $b = \gamma$, unde quinque assertiones in art. praec. propositae sponte demanant.

17.

Disquisitio itaque de indole summae seriei $F(\alpha, \beta, \gamma, 1)$ natura sua restringitur ad casum, quo $\gamma - \alpha - \beta$ est quantitas positiva, ubi illa summa semper exhibet quantitatem finitam. Praemittimus autem observationem sequentem. Si coefficientes seriei $1 + a\omega + b\omega^2 + c\omega^3+$ etc. $= S$ inde a certo termino ultra omnes limites decrescunt, productum

$$(1-\omega)S = 1 + (a-1)\omega + (b-a)\omega^2 + (c-b)\omega^3 + \text{etc.}$$

pro $\omega = 1$ statuere oportet $= 0$, etiamsi summa ipsius seriei S infinite magna evadat. Quoniam enim collectis duobus terminis summa fit $= a$, collectis tribus $= b$, collectis quatuor $= c$ etc., limes summae in infinitum continuatae est $= 0$.

Quoties itaque $\gamma-\alpha-\beta$ est quantitas positiva, statuere oportebit $(1-x)F(\alpha,\beta,\gamma-1,x)=0$ pro $x=1$, unde per aequationem 15 art. 7 habebimus

$$0=\gamma(\alpha+\beta-\gamma)F(\alpha,\beta,\gamma,1)+(\gamma-\alpha)(\gamma-\beta)F(\alpha,\beta,\gamma+1,1), \quad \text{sive}$$

$$F(\alpha,\beta,\gamma,1)=\frac{(\gamma-\alpha)(\gamma-\beta)}{\gamma(\gamma-\alpha-\beta)}\cdot F(\alpha,\beta,\gamma+1,1). \tag{38}$$

Quare quum perinde fiat

$$F(\alpha,\beta,\gamma+1,1)=\frac{(\gamma+1-\alpha)(\gamma+1-\beta)}{(\gamma+1)(\gamma+1-\alpha-\beta)}\cdot F(\alpha,\beta,\gamma+2,1)$$

et sic porro, erit generaliter, k denotante integrum positivum quemcunque

$$\begin{aligned}&F(\alpha,\beta,\gamma,1) \text{ aequalis producto ex } F(\alpha,\beta,\gamma+k,1)\\ &\qquad\text{in } \frac{(\gamma-\alpha)(\gamma+1-\alpha)(\gamma+2-\alpha)\cdot\ldots\cdot(\gamma+k-1-\alpha)}{\gamma(\gamma+1)(\gamma+2)\cdot\ldots\cdot(\gamma+k-1)}\\ &\qquad\text{in } \frac{(\gamma-\beta)(\gamma+1-\beta)(\gamma+2-\beta)\cdot\ldots\cdot(\gamma+k-1-\beta)}{(\gamma-\alpha-\beta)(\gamma+1-\alpha-\beta)(\gamma+2-\alpha-\beta)\cdot\ldots\cdot(\gamma+k-1-\alpha-\beta)}.\end{aligned} \tag{39}$$

18.

Introducamus abhinc sequentem notationem:

$$\Pi(k,z)=\frac{1\cdot2\cdot3\cdot\ldots\cdot k\cdot k^z}{(z+1)(z+2)(z+3)\cdot\ldots\cdot(z+k)} \tag{[art. 38]}$$

ubi k natura sua subintelligitur designare integrum positivum, qua restrictione $\Pi(k,z)$ exhibet functionem duarum quantitatum k,z prorsus determinatam. Hoc modo facile intelligetur, theorema in fine art. praec. propositum ita exhiberi posse

$$F(\alpha,\beta,\gamma,1)=\frac{\Pi(\gamma-\alpha-\beta)\cdot\Pi(k,\gamma)}{\Pi(k,\gamma-\alpha)\cdot\Pi(k,\gamma-\beta)}\cdot F(\alpha,\beta,\gamma+k,1). \tag{[39]}$$

19.

Operae pretium erit, indolem functionis $\Pi(k,z)$ accuratius perpendere. Quoties z est integer negativus, functio manifeste valorem infinite magnum obtinet, simulac ipsi k tribuitur valor satis magnus. Pro valoribus integris ipsius z non negativis autem habemus

$$\begin{aligned}\Pi(k,0)&=1\\ \Pi(k,1)&=k\\ \Pi(k,2)&=\frac{k(k+1)}{1\cdot2}\\ \Pi(k,3)&=\frac{k(k+1)(k+2)}{1\cdot2\cdot3}\\ &\text{etc.}\quad\text{sive generaliter}\end{aligned}$$

$$\Pi(k,z)=\frac{k(k+1)(k+2)\cdot\ldots\cdot(k+z-1)}{1\cdot2\cdot3\cdot\ldots\cdot z} \tag{[40]}$$

Generaliter autem pro quovis valore ipsius z habemus

$$\Pi(k, z+1) = \Pi(k, z) \cdot \frac{k}{z+1}. \tag{[41]}$$

adeoque, quum $\Pi(1, z) = \frac{1}{z+1}$,

$$\Pi(k, z) = \frac{1}{z+1} \cdot \frac{2}{z+2} \cdot \frac{3}{z+3} \cdot \frac{4}{z+4} \cdot \ldots \cdot \frac{k}{z+k}. \tag{[42]}$$

20.

Imprimis vero attentione dignus est limes, ad quem pro valore dato ipsius z functio $\Pi(k, z)$ continuo converget, dum k in infinitum crescit. Sit primo h valor finitus ipsius k maior quam z, patetque, si k transire supponatur ex h in $h+1$, logarithmum ipsius $\Pi(k, z)$ accipere incrementum, quod per seriem convergentem sequentem exprimatur

$$\frac{z}{h+1} - \frac{z(1+z)}{2(h+1)^2} + \frac{z(1+z^2)}{3(h+1)^3} - \frac{z(1-z^2)}{4(h+1)^4} + \text{etc.}$$

Si itaque k e valore h transit in $h+n$, logarithmus ipsius $\Pi(k, z)$ accipiet incrementum

$$\begin{aligned}\frac{nz}{h+1} - \frac{nz(1+z)}{2(h+1)^2} + \frac{nz(1+z^2)}{3(h+1)^3} - \text{etc.}&\\ - \frac{n(n-1)z(1+z)}{2\cdot 2(h+1)^2} + \text{etc.}&\\ + \frac{n(n-1)(n-2)z(1+z^2)}{2\cdot 3\cdot 3(h+1)^3} - \text{etc.}&\\ + \text{etc.}&\end{aligned}$$

quod semper finitum manere, etiamsi n in infinitum crescat, facile demonstrari potest. Quare nisi iam in $\Pi(h, z)$ factor infinitus affuerit, i. e. nisi z sit numerus integer negativus, limes ipsius $\Pi(k, z)$ pro $k = \infty$ certo erit quantitas finita. Manifeste itaque $\Pi(\infty, z)$ tantummodo a z pendet, sive functionem ipsius z ex asse determinatam exhibet, quae abhinc simpliciter per Πz denotabitur. Definimus itaque functionem Πz per valorem producti

$$\frac{1\cdot 2\cdot 3\cdot \ldots \cdot k \cdot k^z}{(z+1)(z+2)(z+3)\cdot \ldots \cdot (z+k)}$$

pro $k = \infty$, aut si mavis per limitem producti infiniti

$$\frac{1^{1+z}}{z+1} \cdot \frac{2^{1+z}}{1^{1+z}(2+z)} \cdot \frac{3^{1+z}}{2^{1+z}(3+z)} \cdot \frac{4^{1+z}}{3^{1+z}(4+z)} \cdots \text{in inf.} \tag{[43]}$$

21.

Ex aequatione 41 statim sequitur aequatio fundamentalis

$$\Pi(z+1) = (z+1)\Pi z \tag{[44]}$$

unde generaliter, designante n integrum positivum quemcunque

$$\Pi(z+n) = (z+1)(z+2)(z+3)\cdot \ldots \cdot (z+n)\Pi z. \tag{[45]}$$

Pro valore integro negative ipsius z erit valor functionis Πz infinite magnus; pro valoribus integris non negativis habemus

$$\begin{array}{ll} \Pi 0 = 1 & \Pi 1 = 1 \\ \Pi 2 = 2 & \Pi 3 = 6 \\ \Pi 4 = 24 & \text{etc.} \end{array}$$

atque generaliter

$$\Pi z = 1 \cdot 2 \cdot 3 \cdot \ldots \cdot z. \tag{[46]}$$

Sed male haec proprietas functionis nostrae tamquam ipsius definitio venditaretur, quippe quae natura sua ad valores integros restringitur, et praeter functionem nostram infinitis aliis (e. g. $\cos 2\pi z \cdot \Pi z$, $\cos^2 \pi z \cdot \Pi z$ etc., denotante π semiperipheriam circuli, cuius radius $= 1$) communis est.

22.

Functio $\Pi(k, z)$, etiamsi generalior videatur quam Πz, tamen abhinc nobis superflua erit, quum facile ad posteriorem reducatur. Colligitur enim e combinatione aequationum 38, 45, 46

$$\Pi(k, z) = \frac{\Pi k \cdot \Pi z}{\Pi(k + z)}. \tag{[47]}$$

Ceterum nexus harum functionum cum iis, quas clar. Kramp facultates numericas nominavit, per se obvius est. Scilicet facultas numerica, quam hic auctor per $a^{c|b}$ designat, in signis nostris est

$$\Pi(\tfrac{a}{b} - 1) \cdot c^{a/b}, \quad \Pi(\tfrac{a-1}{b} - 1) \cdot c^{(a-1)/b}, \quad \Pi(\tfrac{a-2}{b} - 1) \cdot c^{(a-2)/b} \quad \text{etc.}$$

Sed consultius videtur, functionem unius variabilis in analysin introducere, quam functionem trium variabilium, praesertim quum hanc ad illam reducere liceat.

23.

Continuitas functionis Πz interrumpitur, quoties ipsius valor fit infinite magnus, i. e. pro valoribus integris negativis ipsius z. Erit itaque illa positiva a $z = -1$ usque ad $z = \infty$, et quum pro utroque limite Πz obtineat valorem infinite magnum, inter ipsos dabitur valor minimus, quem esse $= 0{,}8856024$ atque respondere valori $z = 0{,}4616321$ invenimus. Inter limites $z = -1$ et $z = -2$, valor functionis Πz fit negativus, inter $z = -2$ atque $z = -3$ iterum positivus et sic porro, uti ex aequ. 44 sponte sequitur. Porro patet, si omnes valores functionis Πz inter limites arbitrarios unitate differentes e. g. a $z = 0$ usque ad $z = 1$ pro notis habere liceat, valorem functionis pro quovis alio valore reali ipsius z adiumento aequationis 45 facile inde deduci posse. Ad hunc finem construximus tabulam, ad calcem huius sectionis annexam, quae ad figuras viginti exhibet logarithmos briggicos functionis Πz, pro $z = 0$ usque ad $z = 1$ per singulas partes centesimas summa cura computatos, ubi tamen monendum, figuram ultimam vigesimam interdum una duabusve unitatibus erroneam esse posse.

24.

Quum limes functionis $F(\alpha, \beta, \gamma + k, 1)$, crescente k in infinitum, manifesto sit unitas, aequatio 39 transit in hanc

$$F(\alpha, \beta, \gamma, 1) = \frac{\Pi(\gamma - \alpha - \beta) \cdot \Pi\gamma}{\Pi(\gamma - \alpha) \cdot \Pi(\gamma - \beta)}. \tag{[48]}$$

quae formula exhibet solutionem completam quaestionis, quae obiectum huius sectionis constituit. Sponte hinc sequuntur aequationes elegantes:

$$F(\alpha, \beta, \gamma, 1) = F(-\alpha, -\beta, \gamma - \alpha - \beta, 1) \tag{[49]}$$
$$F(\alpha, -\beta, \gamma, 1) \cdot F(-\alpha, \beta, \gamma - \alpha, 1) = 1 \tag{[50]}$$
$$F(\alpha, \beta, \gamma, 1) \cdot F(\alpha, -\beta, \gamma - \alpha, 1) = 1 \tag{[51]}$$

in quarum prima γ, in secunda $\gamma - \beta$, in tertia $\gamma - \alpha$ debet esse quantitas positiva.

25.

Applicemus formulam 48 ad quasdam ex aequationibus art. 5. Formula XIII, statuendo $\psi = \frac{1}{2}\pi = \frac{1}{2}\pi$, fit $\psi\pi = F(\frac{1}{2}, \frac{1}{2}, 1, 1)$, sive aequivalet aequationi notae

$$\pi = 1 + \tfrac{1\cdot 1}{2\cdot 2} + \tfrac{1\cdot 1\cdot 3\cdot 3}{2\cdot 2\cdot 4\cdot 4} + \tfrac{1\cdot 1\cdot 3\cdot 3\cdot 5\cdot 5}{2\cdot 2\cdot 4\cdot 4\cdot 6\cdot 6} + \text{etc.}$$

Quare quum per formulam 48 habeatur $F(\frac{1}{2}, \frac{1}{2}, 1, 1) = \frac{\Pi 0\cdot \Pi(-\frac{1}{2})}{\Pi(-\frac{1}{2})^2}$, adeoque, quum $\Pi 0 = 1$, $\Pi\frac{1}{2} = \Pi(-\frac{1}{2})$, fit $\pi = (\Pi(-\frac{1}{2}))^2$ sive:

$$\Pi(-\tfrac{1}{2}) = \sqrt{\pi} \tag{[52]}$$
$$\Pi\tfrac{1}{2} = \sqrt{\pi} \tag{[53]}$$

Formula XVI art. 5, quae aequivalet aequationi notae

$$\sin n\psi = n \sin\psi - \tfrac{n(nn-1)}{1\cdot 2\cdot 3} \sin^3\psi + \tfrac{n(nn-1)(nn-9)}{1\cdot 2\cdot 3\cdot 4\cdot 5} \sin^5\psi - \text{etc.}$$

atque generaliter pro quovis valore ipsius n locum habet, si modo ψ limites -90ř et $+90$ř non transgrediatur, dat pro $t = \sin\psi$

$$F(-\tfrac{1}{2}n, \tfrac{1}{2}n, \tfrac{1}{2}, 1) = \frac{\sin\frac{1}{2}n\pi}{n\sin\frac{1}{2}\pi},$$

unde deducitur formula elegans

$$\Pi\tfrac{1}{2}n \cdot \Pi(-\tfrac{1}{2}n) = \frac{\pi}{\sin\frac{1}{2}n\pi}, \quad \text{sive statuendo } n = 2z$$
$$\Pi z \cdot \Pi(-z) = \frac{\pi}{\sin z\pi} \tag{[54]}$$
$$\Pi(-z) \cdot \Pi z = \frac{\pi z}{\sin z\pi} \tag{[55]}$$

nec non scribendo $z + \frac{1}{2}$ pro z

$$\Pi(-\tfrac{1}{2} + z) \cdot \Pi(-\tfrac{1}{2} - z) = \frac{\pi}{\cos z\pi}. \tag{[56]}$$

E combinatione formulae 54 cum definitione functionis Π sequitur, $\frac{\sin z\pi}{\pi z}$ esse limitem producti infiniti

$$\frac{(1\cdot 2\cdot 3\cdot 4\cdot \ldots \cdot k)^z}{(1 - zz)(4 - zz)(9 - zz)\cdot \ldots \cdot (kk - zz)}$$

crescente k in infinitum, adeoque

$$\sin z\pi = z\pi\Big(1 - \tfrac{zz}{1}\Big)\Big(1 - \tfrac{zz}{4}\Big)\Big(1 - \tfrac{zz}{9}\Big) \text{ etc. in inf.}$$

similique modo ex 56 deducitur

$$\cos z\pi = \Big(1 - 4zz\Big)\Big(1 - \tfrac{4zz}{9}\Big)\Big(1 - \tfrac{4zz}{25}\Big) \text{ etc. in inf.}$$

formulae notissimae, quae ab analystis per methodos prorsus diversas erui solent.

26.

Designante n numerum integrum, valor expressionis

$$\frac{n^{nz}\Pi(k,z)\cdot\Pi(k,z-\frac{1}{n})\cdot\Pi(k,z-\frac{2}{n})\cdot\ldots\cdot\Pi(k,z-\frac{n-1}{n})}{\Pi(nk,nz)}$$

rite collectus invenitur

$$\frac{(1\cdot2\cdot3\cdot\ldots\cdot k)^{n-1}\cdot n^{nk-1}}{1\cdot2\cdot3\cdot\ldots\cdot(nk-1)\cdot(nk)^{nk-1}}$$

adeoque a z est independens, sive idem manebit, quicunque valor ipsi z tribuatur. Exhiberi poterit itaque, quoniam $\Pi(k,0)=\Pi(nk,0)=1$, per productum

$$\Pi(k,-\tfrac{1}{n})\cdot\Pi(k,-\tfrac{2}{n})\cdot\Pi(k,-\tfrac{3}{n})\cdot\ldots\cdot\Pi(k,-\tfrac{n-1}{n}).$$

Crescente igitur k in infinitum, nanciscimur

$$\frac{n^{nz}\Pi z\cdot\Pi(z-\frac{1}{n})\cdot\Pi(z-\frac{2}{n})\cdot\ldots\cdot\Pi(z-\frac{n-1}{n})}{\Pi nz} = \Pi(-\tfrac{1}{n})\cdot\Pi(-\tfrac{2}{n})\cdot\Pi(-\tfrac{3}{n})\cdot\ldots\cdot\Pi(-\tfrac{n-1}{n}). \quad ([57])$$

Productum ad dextram, in se ipsum ordine factorum inverso multiplicatum, producit, per form. 55,

$$\frac{\pi}{\sin\frac{\pi}{n}}\cdot\frac{\pi}{\sin\frac{2\pi}{n}}\cdot\frac{\pi}{\sin\frac{3\pi}{n}}\cdot\ldots\cdot\frac{\pi}{\sin\frac{(n-1)\pi}{n}} = \frac{(2\pi)^{\frac{1}{2}(n-1)}}{\sqrt{n}}$$

Unde habemus theorema elegans

$$\frac{n^{nz}\Pi z\cdot\Pi(z-\frac{1}{n})\cdot\Pi(z-\frac{2}{n})\cdot\ldots\cdot\Pi(z-\frac{n-1}{n})}{\Pi nz} = \frac{(2\pi)^{\frac{1}{2}(n-1)}}{\sqrt{n}}. \qquad ([57])$$

27.

Integrale $\int x^{\lambda-1}(1-x^\mu)^\nu dx$, ita acceptum, ut evanescat pro $x=0$, exprimitur per seriem sequentem, siquidem λ, μ, ν sunt quantitates positivae:

$$\frac{x^\lambda}{\lambda} - \frac{\nu}{1}\cdot\frac{x^{\lambda+\mu}}{\lambda+\mu} + \frac{\nu(\nu-1)}{1\cdot2}\cdot\frac{x^{\lambda+2\mu}}{\lambda+2\mu} - \frac{\nu(\nu-1)(\nu-2)}{1\cdot2\cdot3}\cdot\frac{x^{\lambda+3\mu}}{\lambda+3\mu} + \text{etc.}$$

Hinc ipsius valor pro $x=1$ erit

$$\frac{\Pi(\frac{\lambda}{\mu})\cdot\Pi\nu}{\lambda\cdot\Pi(\frac{\lambda}{\mu}+\nu)}.$$

Ex hoc theoremate omnes relationes, quas ill. Euler olim multo labore evolvit, sponte demanant. Ita e. g. statuendo

$$\int\frac{dx}{\sqrt{(1-x^4)}} = A, \quad \int\frac{xx\,dx}{\sqrt{(1-x^4)}} = B$$

erit $A = \frac{1}{4}\Pi\frac{1}{4}\cdot\Pi(-\frac{1}{2})$, $B = \frac{1}{4}\Pi\frac{3}{4}\cdot\Pi(-\frac{1}{2})$, unde $A\cdot B = \frac{\pi}{4}$. Tum hinc sequitur, quoniam $\Pi\frac{1}{4}\cdot\Pi(-\frac{3}{4}) = \frac{\pi}{\sin\frac{1}{4}\pi} = \pi\sqrt{2}$,

$$\Pi\tfrac{1}{4} = A\cdot\sqrt{\tfrac{\pi}{2B}},\quad \Pi(-\tfrac{3}{4}) = B\cdot\sqrt{\tfrac{2\pi}{A}},\quad \Pi\tfrac{3}{4} = B\cdot\sqrt{\tfrac{\pi}{2A}}.$$

Valor numericus ipsius A, computante Stirling, habetur = 1,31102877714605987, valor ipsius B, secundum eundem auctorem, = 0,59907011736779611, ex nostro calculo, artificio peculiari innixo, = 0,5990701173677961037 2.

Generaliter facile ostendi potest, valorem functionis Πz, si z sit quantitas rationalis $= \frac{m}{\mu}$, denotantibus m, μ integros, ex $(\mu-1)$ valoribus determinatis talium integralium pro $x = 1$ deduci posse, et quidem permultis modis diversis. Accipiendo enim pro λ numerum integrum atque pro ν fractionem, cuius denominator $= \mu$, valor illius integralis semper reducitur ad tres Πz, ubi z est fractio cum denominatore $= \mu$; quodvis vero huiusmodi Πz vel ad $\Pi(\frac{\rho}{\mu})$, vel ad $\Pi(-\frac{\rho}{\mu})$, vel ad $\Pi(\frac{\rho-1}{\mu})$ etc. vel ad $\Pi(-\frac{\rho-1}{\mu})$ reduci potest per formulam 45, siquidem z revera est fractio; si enim z est integer, Πz per se constat. Ex illis vero integralium valoribus, generaliter loquendo, quodvis $\Pi(-\frac{m}{\mu})$, si $m < \frac{1}{2}\mu$, per eliminationem erui potest*).

Quin adeo semissis talium integralium sufficiet, si formulam 54 simul in auxilium vocamus. Ita e. g. statuendo

$$\int\frac{x^{\rho-1}\,dx}{\sqrt[\rho]{(1-x^\rho)}} = C,\quad \int\frac{x^{2\rho-1}\,dx}{\sqrt[\rho]{(1-x^\rho)}} = D,\quad \int\frac{x^{3\rho-1}\,dx}{\sqrt[\rho]{(1-x^\rho)}} = E,\quad \int\frac{x^{4\rho-1}\,dx}{\sqrt[\rho]{(1-x^\rho)}} = F,$$

erit

$$C = \tfrac{1}{\rho}\Pi\tfrac{1}{\rho}\cdot\Pi(-\tfrac{1}{\rho}),\quad D = \tfrac{1}{\rho}\Pi\tfrac{2}{\rho}\cdot\Pi(-\tfrac{1}{\rho}),\quad E = \tfrac{1}{\rho}\Pi\tfrac{3}{\rho}\cdot\Pi(-\tfrac{1}{\rho}),\quad F = \tfrac{1}{\rho}\Pi\tfrac{4}{\rho}\cdot\Pi(-\tfrac{1}{\rho}).$$

Hinc propter $\Pi\frac{1}{\rho} = \frac{\pi}{\sin\frac{\pi}{\rho}}\cdot\Pi(-\frac{\rho-1}{\rho})$, habemus

$$\begin{aligned}\Pi(-\tfrac{1}{\rho}) &= \sqrt{(\rho\cdot CDEF\ldots)},\\ \Pi(-\tfrac{2}{\rho}) &= \sqrt{(\rho\cdot CDE\ldots F^{-1})},\\ \Pi(-\tfrac{3}{\rho}) &= \sqrt{(\rho\cdot CD\ldots E^{-1}F^{-1})}\text{ etc.},\\ \Pi(-\tfrac{\rho-1}{\rho}) &= \sqrt{(\rho\cdot C\cdot D^{-1}E^{-1}F^{-1}\ldots)}.\end{aligned}$$

Formulae 54, 55 adhuc suppeditant

$$C = \frac{\pi}{\rho\sin\frac{\pi}{\rho}},\quad D = \frac{\pi}{\rho\sin\frac{2\pi}{\rho}}\cdot F\quad\text{etc.}$$

ita ut duo integralia D, E vel E et F sufficiant, ad omnes valores $\Pi(-\frac{1}{\rho})$, $\Pi(-\frac{2}{\rho})$ etc. computandos.

*) Haec eliminatio, si pro quantitatibus ipsis logarithmos introducimus, aequationibus tantummodo linearibus applicanda erit.

28.

Statuendo $y = x^{1/\nu}$, atque $\frac{1}{\nu} = \lambda$, $\frac{\mu}{\nu} = \beta$, erit valor integralis $\int y^{\alpha\lambda+\lambda-1}(1-y^\lambda)^{\nu\beta}dy$ ab $y = 0$ usque ad $y = 1$, sive valor integralis $\int x^{\alpha-1}(1-\frac{x}{\nu})^{\nu\beta}dx$ inter eosdem limites $= \frac{\Pi(\alpha-1)\cdot\Pi(\nu\beta)}{\lambda\cdot\Pi(\alpha+\nu\beta-1)} = \frac{\Pi(\alpha-1)}{\beta}$ (form. 47), siquidem ν denotet integrum. Iam crescente ν in infinitum, limes ipsius $\Pi(\nu,\lambda)$ erit $=\Pi\lambda$, limes ipsius $(1-\frac{x}{\nu})^{\nu\beta}$ autem $e^{-\beta x}$, denotante e basin logarithmorum hyperbolicorum. Quamobrem si λ est positiva, $\frac{\Pi(\alpha-1)}{\beta}$ sive $\Pi(\lambda-1)$ exprimet integrale $\int y^{\lambda-1}e^{-y}dy$ ab $y=0$ usque ad $y=\infty$, sive scribendo λ pro $\lambda-1$, $\Pi\lambda$ est valor integralis $\int y^{\lambda}e^{-y}dy$ ab $y=0$ usque ad $y=\infty$, si $\lambda+1$ est quantitas positiva.

Generalius statuendo $y=z^{\alpha}$, $\alpha\lambda+\alpha-1=\beta$, transit $\int y^{\lambda}e^{-y}dy$ in $\alpha\int z^{\beta}e^{-z^{\alpha}}dz$, quod itaque inter limites $z=0$ atque $z=\infty$ sumtum exprimitur per $\Pi(\frac{\beta+1}{\alpha}-1)$ sive

$$\text{Valor integralis } \int z^{\beta}e^{-z^{\alpha}}dz, \text{ a } z=0 \text{ usque ad } z=\infty \text{ fit} = \frac{\Pi(\frac{\beta+1}{\alpha}-1)}{\alpha} = \frac{\Pi\frac{\beta+1}{\alpha}}{(\beta+1)}. \qquad \text{([art. 28])}$$

Si modo α atque $\beta+1$ sunt quantitates positivae (si utraque est negativa, integrale per $\frac{\Pi(\frac{\beta+1}{\alpha}-1)}{\alpha}$ exprimetur). Ita e. g. pro $\beta=0$, $\alpha=2$, valor integralis $\int e^{-zz}dz$ invenitur $=\frac{\Pi\frac{1}{2}}{2}=\frac{1}{2}\sqrt{\pi}$.

29.

Ill. Euler pro summa logarithmorum $\log 1+\log 2+\log 3+$ etc. $+\log z$ eruit seriem $(z+\frac{1}{2})\log z-z+\frac{1}{2}\log 2\pi+\frac{\mathfrak{A}}{2z}-\frac{\mathfrak{B}}{24z^3}+\frac{\mathfrak{C}}{120z^5}-\frac{\mathfrak{D}}{168z^7}+$ etc. ubi $\mathfrak{A}=\frac{1}{6}$, $\mathfrak{B}=\frac{1}{30}$, $\mathfrak{C}=\frac{1}{42}$ etc. sunt numeri Bernoulliani. Per hanc itaque seriem exprimitur $\log\Pi z$; etiamsi enim primo aspectu haec conclusio ad valores integros restricta videatur, tamen rem propius contemplando invenietur, evolutionem ab Eulero adhibitam (Instit. Calc. Diff. Cap. vi. 159) saltem ad valores positivos fractos eodem iure applicari posse, quo ad integros: supponit enim tantummodo, functionem ipsius z, in seriem evolvendam, esse talem, ut ipsius diminutio, si z transeat in $z-1$, exhiberi possit per theorema Taylori, simulque ut eadem diminutio sit $=\log z$. Conditio prior innititur continuitati functionis, adeoque locum non habet pro valoribus negativis ipsius z, ad quos proin seriem illam extendere non licet: conditio posterior autem functioni $\log\Pi z$ generaliter competit sine restrictione ad valores integros ipsius z. Statuemus itaque

$$\log\Pi z=(z+\tfrac{1}{2})\log z-z+\tfrac{1}{2}\log 2\pi+\frac{\mathfrak{A}}{2z}-\frac{\mathfrak{B}}{24z^3}+\frac{\mathfrak{C}}{120z^5}-\frac{\mathfrak{D}}{168z^7}+\text{etc.} \qquad \text{([58])}$$

Quum hinc quoque habetur

$$\log\Pi 2z=(2z+\tfrac{1}{2})\log 2z-2z+\tfrac{1}{2}\log 2\pi+\frac{\mathfrak{A}}{4z}-\frac{\mathfrak{B}}{96z^3}+\frac{\mathfrak{C}}{480z^5}-\frac{\mathfrak{D}}{672z^7}+\frac{\mathfrak{E}}{1536z^9}-\text{etc.}$$

atque per formulam 57, statuendo $n=2$,

$$\log\Pi(z-\tfrac{1}{2})=\log\Pi 2z-\log\Pi z-(2z+\tfrac{1}{2})\log 2+\tfrac{1}{2}\log 2\pi,$$

fit

$$\log\Pi(z-\tfrac{1}{2})=z\log z-z+\tfrac{1}{2}\log 2\pi-\frac{\mathfrak{A}}{2\cdot 2z}+\frac{\mathfrak{B}}{3\cdot 4\cdot 8z^3}-\frac{\mathfrak{C}}{5\cdot 6\cdot 32z^5}+\frac{\mathfrak{D}}{7\cdot 8\cdot 128z^7}-\text{etc.} \qquad \text{([59])}$$

Hae duae series pro valoribus magnis ipsius z ab initio satis promte convergunt, ita ut summam approximatam commode satisque exacte colligere liceat: attamen probe notandum est, pro quovis valore dato ipsius z, quantumvis magno, praecisionem limitatam tantummodo obtineri posse, quum numeri Bernoulliani seriem hypergeometricam constituant, adeoque series illae, si modo satis longo extendantur, certo e convergentibus divergentes evadant. Ceterum negari nequit, theoriam talium serierum divergentium adhuc quibusdam difficultatibus premi, de quibus forsan alia occasione pluribus commentabimur.

30.

E formula 38 sequitur

$$\frac{\Pi(k,z+\omega)}{\Pi(k,z)} = \frac{z+1}{z+1+\omega} \cdot \frac{z+2}{z+2+\omega} \cdot \frac{z+3}{z+3+\omega} \cdot \ldots \cdot \frac{z+k}{z+k+\omega}$$

unde sumtis logarithmis, in series infinitas evolutis, prodit

$$\begin{aligned}\log\Pi(k,z+\omega) = \log\Pi(k,z) + \omega\Big(&\frac{1}{z+1} + \frac{1}{z+2} + \frac{1}{z+3} + \ldots + \frac{1}{z+k}\\ &- \log(k+1)\Big) + \frac{\omega^2}{2}\Big(\frac{1}{(z+1)^2} + \frac{1}{(z+2)^2} + \ldots + \frac{1}{(z+k)^2}\Big)\\ &- \frac{\omega^3}{3}\Big(\frac{1}{(z+1)^3} + \frac{1}{(z+2)^3} + \ldots + \frac{1}{(z+k)^3}\Big) + \text{etc. in inf.}\end{aligned} \qquad ([60])$$

Series, hic in ω multiplicata, quae, si magis placet, ita etiam exhiberi potest,

$$\frac{1}{z+1} + \log\frac{2}{z+2} + \log\frac{3}{z+3} + \log\frac{4}{z+4} + \ldots + \log\frac{k}{z+k}$$

e terminorum multitudine finita constat, crescente autem k in infinitum, ad limitem certum converget, qui novam functionum transscendentium speciem nobis sistit, in posterum per Ψz denotandam.

Designando porro summas serierum sequentium, in infinitum extensarum,

$$\begin{aligned}&\frac{1}{(z+1)^2} + \frac{1}{(z+2)^2} + \frac{1}{(z+3)^2} + \text{etc.}\\ &\frac{1}{(z+1)^3} + \frac{1}{(z+2)^3} + \frac{1}{(z+3)^3} + \text{etc.}\\ &\frac{1}{(z+1)^4} + \frac{1}{(z+2)^4} + \frac{1}{(z+3)^4} + \text{etc.}\\ &\qquad\qquad\qquad\qquad\qquad\qquad\text{etc.}\end{aligned}$$

resp. per P, Q, R etc. (pro quibus signa functionalia introducere minus necessarium videtur), habebimus

$$\log\Pi(z+\omega) = \log\Pi z + \omega\Psi z + \tfrac{1}{2}\omega^2 P - \tfrac{1}{3}\omega^3 Q + \tfrac{1}{4}\omega^4 R - \text{etc.} \qquad ([61])$$

Manifesto functio Ψz erit functio derivata prima functionis $\log\Pi z$, adeoque

$$\frac{d\Pi z}{dz} = \Pi z \cdot \Psi z. \qquad ([62])$$

Perinde erit $P = \frac{d\Psi z}{dz}$, $Q = \frac{1}{2}\frac{dP}{dz}$, $R = \frac{1}{3}\frac{dQ}{dz}$ etc.

31.

Functio Ψz aeque fere memorabilis est atque functio Πz, quapropter insigniores relationes ad illam spectantes hic colligemus. E differentiatione aequationis 44 fit

$$\Psi(z+1) = \Psi z + \frac{1}{z+1} \qquad ([63])$$

unde

$$\Psi(z+n) = \Psi z + \frac{1}{z+1} + \frac{1}{z+2} + \frac{1}{z+3} + \ldots + \frac{1}{z+n}. \qquad ([64])$$

Huius adiumento a valoribus minoribus ipsius z ad maiores progredi, vel a maioribus ad minores regredi licet: pro valoribus maioribus positivis ipsius z functionis valores numerici satis commode per formulas sequentes e differentiatione aequationum 58, 59 oriundas computantur, de quibus tamen eadem sunt tenenda, quae in art. 29 circa formulas 58 et 59 monuimus:

$$\Psi z = \log z + \frac{\mathfrak{A}}{2zz} - \frac{\mathfrak{B}}{4z^4} + \frac{\mathfrak{C}}{6z^6} - \text{etc.} \qquad ([65])$$

$$\Psi(z-\tfrac{1}{2}) = \log z + \frac{\mathfrak{A}}{2\cdot 2zz} - \frac{\mathfrak{B}}{4\cdot 8z^4} + \frac{\mathfrak{C}}{6\cdot 32z^6} - \text{etc.} \qquad ([66])$$

Ita pro $z = 10$ computavimus

$$\Psi 10 = 2{,}35175258906672110764743$$

unde regredimur ad

$$\Psi 0 = -0{,}57721566490153286060653^{*}).$$

Pro valore integro positivo ipsius z fit generaliter

$$\Psi z = \Psi 0 + 1 + \tfrac{1}{2} + \tfrac{1}{3} + \text{etc.} + \tfrac{1}{z}. \qquad ([67])$$

Pro valore integro negative autem manifesto Ψz fit quantitas infinite magna.

*) Quam hic valor inde a figura vigesima discrepet ab eo quem computavit clar. Mascheroni in Adnotat. ad Euleri Calculum Integr., adhortatus sum Friedericum Bernhardum Gothofredum Nicolai, iuvenem in calculo indefessum, ut computum illum repeteret ulteriusque extenderet. Invenit itaque per calculum duplicem, scilicet descendens tum a $z = 50$ tum a $z = 100$,

$$\Psi 0 = -0{,}5772156649015328606512008240243104 21.$$

Eidem calculatori exercitatissimo etiam debetur tabulae ad finem huius Sectionis annexae pars altera, exhibens valores functionis Ψz ad 18 figuras (quarum ultima haud certa), pro omnibus valoribus ipsius z a 0 usque ad 1 per singulas partes centesimas. Ceterum methodi, per quas utraque tabula constructa est, innituntur partim theorematibus quae hic traduntur, partim calculi artificiis singularibus, quae alia occasione proferemus.

32.

Formula 55 nobis suppeditat $\log \Pi(-z) + \log \Pi(z-1) = \log \pi - \log \sin z\pi$, unde fit per differentiationem

$$\Psi(-z) - \Psi(z-1) = \pi z \pi. \qquad ([68])$$

Et quam e definitione functionis Ψ generaliter habeatur

$$\Psi x - \Psi y = \frac{1}{x+1} - \frac{1}{y+1} + \frac{1}{x+2} - \frac{1}{y+2} + \frac{1}{x+3} - \frac{1}{y+3} + \text{etc.} \qquad ([69])$$

oritur series nota

$$\pi z \pi = \frac{1}{z} - \frac{1}{1-z} + \frac{1}{1+z} - \frac{1}{2-z} + \frac{1}{2+z} - \frac{1}{3-z} + \text{etc.}$$

Simili modo e differentiatione formulae 57 prodit

$$\Psi z + \Psi(z - \tfrac{1}{n}) + \Psi(z - \tfrac{2}{n}) + \ldots + \Psi(z - \tfrac{n-1}{n}) = n\Psi nz - n\log n. \qquad ([70])$$

adeoque statuendo $z = 0$

$$\Psi(-\tfrac{1}{n}) + \Psi(-\tfrac{2}{n}) + \Psi(-\tfrac{3}{n}) + \ldots + \Psi(-\tfrac{n-1}{n}) = (n-1)\Psi 0 - n\log n. \qquad ([71])$$

Ita e. g. habetur

$$\begin{aligned}\Psi(-\tfrac{1}{2}) &= \Psi 0 - 2\log 2 = -1{,}9635100260214234794409 9, \text{ unde porro}\\ \Psi\tfrac{1}{2} &= +0{,}0364899739785765205590 1.\end{aligned}$$

33.

Sicuti in art. praec. $\Psi(-\frac{1}{2})$ ad $\Psi 0$ et logarithmum reduximus, ita generaliter $\Psi(-\frac{m}{n})$, designantibus m, n integros, quorum minor m, ad $\Psi 0$ et logarithmos reducemus. Statuamus $\frac{\pi}{n} = \omega$, sitque φ alicui angulorum ω, 2ω, 3ω, …, $(n-1)\omega$ aequalis; unde $1 = \cos n\varphi = \cos 2n\varphi = \cos 3n\varphi$ etc., $\cos\varphi = \cos(n+1)\varphi = \cos(n+2)\varphi$ etc., $\cos 2\varphi = \cos(n+2)\varphi$ etc., nec non $\cos\varphi + \cos 2\varphi + \cos 3\varphi +$ etc. $+ \cos(n-1)\varphi + 1 = 0$. Habemus itaque

$$\begin{aligned}\cos\varphi\cdot\Psi\tfrac{1}{n} &= -\tfrac{1}{n}\cos\varphi + \cos\varphi\cdot\log 2 - \tfrac{1}{2}\cos(n+1)\varphi + \cos\varphi\cdot\log\tfrac{1}{2} - \text{etc.}\\ \cos 2\varphi\cdot\Psi\tfrac{2}{n} &= -\tfrac{1}{n}\cos 2\varphi + \cos 2\varphi\cdot\log 2 - \tfrac{1}{2}\cos(n+2)\varphi + \cos 2\varphi\cdot\log\tfrac{1}{2} - \text{etc.}\\ \cos 3\varphi\cdot\Psi\tfrac{3}{n} &= -\tfrac{1}{n}\cos 3\varphi + \cos 3\varphi\cdot\log 2 - \tfrac{1}{2}\cos(n+3)\varphi + \cos 3\varphi\cdot\log\tfrac{1}{2} - \text{etc.}\\ &\text{etc. usque ad}\\ \cos(n-1)\varphi\cdot\Psi(\tfrac{n-1}{n}) &= -\tfrac{1}{n}\cos(n-1)\varphi + \cos(n-1)\varphi\cdot\log 2 - \tfrac{1}{2}\cos(2n-1)\varphi\\ &\quad + \cos(n-1)\varphi\cdot\log\tfrac{1}{2} - \text{etc.}\end{aligned}$$

atque per summationem

$$\begin{aligned}&\cos\varphi\cdot\Psi\tfrac{1}{n} + \cos 2\varphi\cdot\Psi\tfrac{2}{n} + \cos 3\varphi\cdot\Psi\tfrac{3}{n} + \text{etc.} + \cos(n-1)\varphi\cdot\Psi(\tfrac{n-1}{n}) + \Psi 0\\ &\qquad = -\tfrac{1}{n}(\cos\varphi + \tfrac{1}{2}\cos 2\varphi + \tfrac{1}{3}\cos 3\varphi + \tfrac{1}{4}\cos 4\varphi + \text{etc. in infin.})\end{aligned}$$

Sed habetur generaliter, pro valore ipsius x unitate non maiori,

$$\log(1 - 2x\cos\varphi + xx) = -2(x\cos\varphi + \tfrac{1}{2}xx\cos 2\varphi + \tfrac{1}{3}x^3\cos 3\varphi + \text{etc.})$$

quae quidem series facile sequitur ex evolutione $\log(1 - rx) + \log(1 - \frac{x}{r})$, denotante r quantitatem $\cos\varphi + \sqrt{-1}\cdot\sin\varphi$. Hinc fit aequatio praecedens

$$\begin{aligned}&\cos\varphi\cdot\Psi\tfrac{1}{n} + \cos 2\varphi\cdot\Psi\tfrac{2}{n} + \cos 3\varphi\cdot\Psi\tfrac{3}{n} + \text{etc.} + \cos(n-1)\varphi\cdot\Psi(\tfrac{n-1}{n})\\ &\qquad = -\Psi 0 + \tfrac{1}{2n}\log(2 - 2\cos\varphi). \qquad ([72])\end{aligned}$$

Statuatur in hac aequatione deinceps $\varphi = \omega$, $\varphi = 2\omega$, $\varphi = 3\omega$ etc. usque ad $\varphi = (n-1)\omega$, multiplicentur singulae hae aequationes ordine suo per $\cos m\omega$, $\cos 2m\omega$, $\cos 3m\omega$ etc. usque ad $\cos(n-1)m\omega$, productorumque aggregato adiiciatur aequatio 71

$$\Psi 0 = -\tfrac{1}{n} - \tfrac{1}{n}\cos m\omega + \log 2 - \tfrac{1}{2}\cos 2m\omega + \log\tfrac{1}{2} - \text{etc.}$$

Quodsi iam perpenditur, esse

$$1 + \cos m\omega \cdot \cos k\omega + \cos 2m\omega \cdot \cos 2k\omega + \cos 3m\omega \cdot \cos 3k\omega \\ + \text{etc.} + \cos(n-1)m\omega \cdot \cos(n-1)k\omega = 0$$

denotante k aliquem numerorum 1, 2, 3, ..., $(n-1)$ exceptis his duobus m atque $n-m$, pro quibus summa illa fit $= \frac{1}{2}n$, patebit, ex summatione illarum aequationum prodire, post divisionem per $\frac{1}{2}n$,

$$\Psi(-\tfrac{m}{n}) = \Psi 0 + \tfrac{1}{2}\pi\tfrac{m\pi}{n} - \log n + \cos\tfrac{m\pi}{n}\cdot\log(2-2\cos\tfrac{\pi}{n}) \\ + \cos\tfrac{2m\pi}{n}\cdot\log(2-2\cos\tfrac{2\pi}{n}) + \cos\tfrac{3m\pi}{n}\cdot\log(2-2\cos\tfrac{3\pi}{n}) \\ + \text{etc.} + \cos\tfrac{(n-1)m\pi}{n}\cdot\log(2-2\cos\tfrac{(n-1)\pi}{n}). \quad ([73])$$

Manifesto terminus ultimus huius aequationis fit $= \cos m\pi \cdot \log(2-2\cos\pi)$, penultimus $= \cos 2m\omega \cdot \log(2-2\cos 2\omega)$ etc., ita ut bini termini semper sint aequales, excepto, si n est par, termino singulari $\cos\frac{n}{2}m\omega \cdot \log(2-2\cos\frac{n}{2}\omega)$, qui fit $= +2\log 2$ pro m pari, vel $= -2\log 2$ pro m impari.

Combinando iam cum aequatione 73 hanc

$$\Psi(-\tfrac{m}{n}) - \Psi(\tfrac{m-n}{n}) = \pi\tfrac{m\pi}{n}$$

habemus, pro valore impari ipsius n, siquidem m est integer positivus minor quam n

$$\Psi(-\tfrac{m}{n}) = \Psi 0 + \tfrac{1}{2}\pi\tfrac{m\pi}{n} - \log n + \cos\tfrac{m\pi}{n}\cdot\log(2-2\cos\tfrac{\pi}{n}) \\ + \cos\tfrac{2m\pi}{n}\cdot\log(2-2\cos\tfrac{2\pi}{n}) + \cos\tfrac{3m\pi}{n}\cdot\log(2-2\cos\tfrac{3\pi}{n}) \\ + \text{etc.} + \cos\tfrac{(n-1)m\pi}{n}\cdot\log(2-2\cos\tfrac{(n-1)\pi}{n}). \quad ([74])$$

Pro valore pari ipsius n autem

$$\Psi(-\tfrac{m}{n}) = \Psi 0 + \tfrac{1}{2}\pi\tfrac{m\pi}{n} - \log n + \cos\tfrac{m\pi}{n}\cdot\log(2-2\cos\tfrac{\pi}{n}) \\ + \cos\tfrac{2m\pi}{n}\cdot\log(2-2\cos\tfrac{2\pi}{n}) + \text{etc.} + \cos\tfrac{(n-1)m\pi}{n}\cdot\log(2-2\cos\tfrac{(n-1)\pi}{n}) \\ \pm \log 2 \quad ([75])$$

ubi signum superius valet pro m pari, inferius pro impari. Ita e. g. invenitur

$$\Psi(-\tfrac{1}{4}) = \Psi 0 + \tfrac{1}{2}\pi - 3\log 2, \qquad \Psi(-\tfrac{3}{4}) = \Psi 0 - \tfrac{1}{2}\pi - 3\log 2,$$
$$\Psi(-\tfrac{1}{3}) = \Psi 0 + \tfrac{1}{6}\pi\sqrt{3} - \tfrac{3}{2}\log 3, \qquad \Psi(-\tfrac{2}{3}) = \Psi 0 - \tfrac{1}{6}\pi\sqrt{3} - \tfrac{3}{2}\log 3.$$

Ceterum combinatis bis aequationibus cum aequatione 64 sponte patet, Ψz generaliter pro quovis valore rationali ipsius z, positivo seu negativo per $\Psi 0$ atque logarithmos determinari posse, quod theorema sane maxime est memorabile.

34.

Quum, per art. 28, $\Pi\lambda$ sit valor integralis $\int y^\lambda e^{-y} dy$, ab $y = 0$ usque ad $y = \infty$, siquidem $\lambda + 1$ est quantitas positiva, fit differentiando secundum λ

$$\frac{d\Pi\lambda}{d\lambda} = \int y^\lambda e^{-y} \log y \cdot dy = \Pi\lambda \cdot \Psi\lambda$$

sive

$$\Pi\lambda \cdot \Psi\lambda = \int y^\lambda e^{-y} \log y \cdot dy, \quad \text{ab } y = 0 \text{ usque ad } y = \infty. \qquad ([76])$$

Generalius statuendo $y = z^\alpha$, $\alpha\lambda + \alpha - 1 = \beta$, valor integralis $\int z^\beta e^{-z^\alpha} \log z \, dz$, a $z = 0$ usque ad $z = \infty$, fit

$$\frac{\alpha\alpha}{\beta + 1} \cdot \frac{\Pi(\frac{\beta+1}{\alpha} - 1)}{\alpha} \cdot \Psi(\tfrac{\beta+1}{\alpha} - 1) \quad \text{sive} \quad \frac{\alpha}{\beta + 1} \cdot \Pi\tfrac{\beta+1}{\alpha} \cdot \Psi(\tfrac{\beta+1}{\alpha} - 1) \qquad ([\text{art. } 34])$$

siquidem simul $\beta + 1$ atque α sunt quantitates positivae, vel aequalis eidem quantitati cum signo opposito, si utraque $\beta + 1$, α est negativa.

35.

At non solum productum $\Pi\lambda \cdot \Psi\lambda$, verum etiam ipsa functio $\Psi\lambda$ per integrale determinatum exhiberi potest. Designante k integrum positivum, patet valorem integralis $\int \frac{1-x^k}{1-x} \, dx$ ab $x = 0$ usque ad $x = 1$ esse

$$= \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \text{etc.} + \frac{1}{k}.$$

Porro quum valor integralis $\int (\frac{1}{x} - \frac{1}{1-x}) dx$ generaliter sit $=$ Const. $+ \log \frac{x}{1-x}$, idem inter limites $x = 0$ atque $x = \frac{1}{k}$ erit $= \log k$, unde patet, valorem integralis

$$S = \int \Big(\frac{1 - (1-x)^k}{x} - \frac{1 - (1-x)^{k-1}}{1-x} \Big) dx$$

inter eosdem limites esse

$$= \log k - \frac{1}{1} - \frac{1}{2} - \frac{1}{3} - \text{etc.} - \frac{1}{k}$$

quam expressionem denotabimus per Q. Discrepamus integrale S in duas partes

$$S = \int \frac{1 - (1-x)^k}{x} \, dx - \int \frac{1 - (1-x)^{k-1}}{1-x} \, dx.$$

Pars prima $\int \frac{1-(1-x)^k}{x} \, dx$, statuendo $x = y^k$ mutatur in

$$\int \frac{1 - (1 - y^k)^k}{y^k} \cdot k y^{k-1} dy = k \int \frac{1 - (1 - y^k)^k}{y} \, dy.$$

unde sponte patet, illius valorem ab $x = 0$ usque ad $x = \frac{1}{k}$, aequalem esse valori integralis

$$\int \frac{k x^{k-1} - k x^{k-1} (1 - x^k)^{k-1}}{x} \, dx$$

inter eosdem limites, quum manifesto literam y sub hac restrictione in x mutare liceat. Hinc fit integrale S, inter eosdem limites

$$= \int \frac{kx^{k-1} - kx^{k-1}(1-x^k)^{k-1}}{x}\,dx - \int \frac{1-(1-x)^{k-1}}{1-x}\,dx.$$

Hoc vero integrale, statuendo $x^k = z$, transit in

$$Q = \int \frac{1-(1-z)^k}{z}\,dz - \int \frac{1-(1-z)^{k-1}}{1-z}\,dz,$$

a $z = 0$ usque ad $z = 1$.

Crescente k in infinitum, limes ipsius Q est $\Psi\lambda$, limes ipsius $\frac{1-(1-z)^k}{z}$ vero $-\log(1-z)$, limes ipsius $k(1-z)^k$ vero est $\log\frac{1}{1-z}$ sive $-\log(1-z)$. Quare habemus

$$\Psi\lambda = \int\Bigl(\frac{1}{1-z} + \frac{1}{\log z}\Bigr) z^{\lambda-1}\,dz, \quad \text{a } z = 0 \text{ usque ad } z = 1. \qquad ([77])$$

36.

Integralia determinata, per quae supra expressae sunt functiones $\Pi\lambda$, $\Pi\lambda\cdot\Psi\lambda$, restringere oportuit ad valores ipsius λ tales, ut $\lambda+1$ evadat quantitas positiva: haec restrictio ex ipsa deductione demanavit, reveraque facile perspicitur, pro aliis valoribus ipsius λ illa integralia semper fieri infinita, etiamsi functiones $\Pi\lambda$, $\Pi\lambda\cdot\Psi\lambda$ finitae manere possint. Veritati formula 77 certo eadem conditio subesse debet, ut $\lambda+1$ sit quantitas positiva (alioquin enim integrale certo infinitum evadit, etiamsi functio $\Psi\lambda$ maneat finita): sed deductio formulae primo aspectu generalis nullique restrictioni obnoxia esse videtur. Sed propius attendenti facile patebit, ipsi analysi, per quam formula eruta est, hanc restrictionem iam inesse. Scilicet tacite supposuimus, integrale $\int\frac{dx}{1-x}$, cui $\int\frac{dx}{x}$ substituimus, habere valorem finitum, quae conditio requirit, ut $\lambda+1$ sit quantitas positiva. Ex analysi nostra quidem sequitur, haec duo integralia semper esse aequalia, si hoc extendatur ab $x=0$ usque ad $x = 1-\omega$, illud ab $x = 0$ usque ad $x = (1-\omega)^\lambda$, quantumvis parva sit quantitas ω, modo non sit $= 0$: sed hoc non obstante in casu eo, ubi $\lambda+1$ non est quantitas positiva, duo integralia ab $x = 0$ usque ad eundem terminum $x = 1-\omega$ extensa neutiquam ad aequalitatem convergunt, sed potius tunc ipsorum differentia, decrescente ω in infinitum, in infinitum crescet. Hocce exemplum monstrat, quanta circumspicientia opus sit in tractandis quantitatibus infinitis, quae in ratiociniis analyticis nostro iudicio eatenus tantum sunt admittendae, quatenus ad theoriam limitum reduci possunt.

37.

Statuendo in formula 77, $z = e^{-u}$, patet, illam etiam ita exhiberi posse

$$\Psi\lambda = -\int\Bigl(\frac{1}{1-e^{-u}} - \frac{1}{u}\Bigr)e^{-\lambda u}\,du, \quad \text{ab } u = \infty \text{ usque ad } u = 0,$$

i. e.

$$\Psi\lambda = \int\Bigl(\frac{1}{1-e^{-u}} - \frac{1}{u}\Bigr)e^{-\lambda u}\,du, \quad \text{ab } u = 0 \text{ usque ad } u = \infty. \qquad ([78])$$

(Perinde valor ipsius $\Pi\lambda$ in art. 28 allatus, mutatur statuendo $e^{-y} = u$, in sequentem

$$\Pi\lambda = \int\Bigl(\log\tfrac{1}{u}\Bigr)^\lambda du, \quad \text{a } u = 0 \text{ usque ad } u = 1.)$$

Porro patet e formula 77, esse

$$\Psi'\lambda - \Psi'\mu = \int \frac{z^{\lambda-1} - z^{\mu-1}}{1-z}\,dz, \quad \text{a } z = 0 \text{ usque ad } z = 1 \tag{[79]}$$

ubi praeter $\lambda + 1$ etiam $\mu + 1$ debet esse quantitas positiva.

Statuendo in eadem formula 77, $z = u^\alpha$, designante α quantitatem positivam, fit

$$\Psi\lambda = \int\Big(\frac{\alpha u^{\alpha-1}}{1-u^\alpha} + \frac{1}{\log u}\Big)u^{\alpha\lambda-1}\,du, \quad \text{ab } u = 0 \text{ usque ad } u = 1$$

et quum perinde statui possit, pro valore positivo ipsius β,

$$\Psi\mu = \int\Big(\frac{\beta u^{\beta-1}}{1-u^\beta} + \frac{1}{\log u}\Big)u^{\beta\mu-1}\,du$$

patet, fieri

$$\int \frac{u^{\alpha-1}}{1-u^\alpha}\,du = \int \frac{u^{\beta-1}}{1-u^\beta}\,du$$

sive

$$\int \frac{u^{\alpha-1}}{1-u^\alpha}\,du = \int \frac{u^{\beta-1}}{1-u^\alpha}\,du + \int \frac{u^{\beta-1}}{1-u^\beta}\,du - \int \frac{u^{\beta-1}}{1-u^\alpha}\,du$$

integralibus semper ab $u = 0$ usque ad $u = 1$ extensis. Sed ponendo $\lambda = 0$, integrale posterius indefinite assignari potest; est scilicet $= \log\frac{\beta}{\alpha}$, si evanescere debet pro $u = 0$; quare quum pro $u = 1$ statuere oporteat $\frac{\beta}{\alpha} = \frac{1}{\alpha}$, erit integrale $\log\frac{1}{\alpha} = \frac{\pi}{\sin\frac{\pi}{\alpha}} - \frac{\pi}{\tan\frac{\pi}{\alpha}}$, ab $u = 0$ usque ad $u = 1$, quod theorema olim ab ill. Euler per alias methodos erutum est.

Methodus Nova

Integralium valores per approximationem inveniendi.[1]

1.

Inter methodos ad determinationem numericam approximatam integralium propositas insignem tenent locum regulae, quas praeeunte summo Newton evolutas dedit Cotes. Scilicet si requiritur valor integralis $\int y\,dx$ ab $x = g$ usque ad $x = h$ sumendus, valores ipsius y pro his valoribus extremis ipsius x et pro quotcunque aliis intermediis a primo ad ultimum incrementis aequalibus progredientibus, multiplicandi sunt per certos coefficientes

[1] Auctore Carolo Friderico Gauss. Societati Regiae Scientiarum exhibita 1814. Sept. 16. Commentationes societatis regiae scientiarum Gottingensis recentiores. Vol. iii. Gottingae MDCCCXVI.

numericos, quo facto productum aggregatum in $h-g$ ductum integrale quaesitum suppeditabit, eo maiore praecisione, quo plures termini in hac operatione adhibentur. Quum principia huius methodi, quae a geometris rarius quam par est in usum vocari videtur, nusquam quod sciam plenius explicata sint, pauca de his praemittere ab instituto nostro haud alienum erit.

2.

Sit $n+1$ multitudo terminorum, quos in usum vocare placuit, statuamusque $h-g=L$, ita ut valores ipsius x sint g, $g+\frac{L}{n}$, $g+\frac{2L}{n}$, $g+\frac{3L}{n}$ etc. usque ad $g+L$, respondeantque iisdem resp. valores ipsius y hi A, A', A'', A''' etc. usque ad $A^{(n)}$; denique ponatur indefinite $x=g+Lt$, ita ut y etiam spectari possit tamquam functio ipsius t. Designemus per Y functionem sequentem

$$\begin{aligned} Y = A\cdot\frac{(nt-1)(nt-2)(nt-3)\cdot\ldots\cdot(nt-n)}{(-1)(-2)(-3)\cdot\ldots\cdot(-n)} & \\ + A'\cdot\frac{nt\cdot(nt-2)(nt-3)\cdot\ldots\cdot(nt-n)}{1\cdot(-1)(-2)\cdot\ldots\cdot(1-n)} & \\ + A''\cdot\frac{nt(nt-1)(nt-3)\cdot\ldots\cdot(nt-n)}{2\cdot 1\cdot(-1)\cdot\ldots\cdot(2-n)} & \\ + A'''\cdot\frac{nt(nt-1)(nt-2)\cdot\ldots\cdot(nt-n)}{3\cdot 2\cdot 1\cdot\ldots\cdot(3-n)} & \\ + \text{etc.} & \\ + A^{(n)}\cdot\frac{nt(nt-1)(nt-2)\cdot\ldots\cdot(nt-n+1)}{n(n-1)(n-2)\cdot\ldots\cdot 1} & \end{aligned}$$

sive $Y=\Sigma$, ubi repraesentante μ singulos integros 0, 1, 2, 3, …, n,

$$\frac{A^{(\mu)}\cdot nt(nt-1)(nt-2)\cdot\ldots\cdot(nt-n)}{\mu!(nt-\mu)(-1)^{n-\mu}(n-\mu)!}$$

$M^{(\mu)}$ valor ipsius Y pro $nt=\mu$.

Manifestum erit, Y exhibere functionem algebraicam integram ipsius t ordinis n, atque eius valores pro singulis $n+1$ valoribus ipsius t, puta 0, $\frac{1}{n}$, $\frac{2}{n}$, $\frac{3}{n}$, …, 1 aequales esse valoribus ipsius y. Porro patet, si Y' sit functio alia integra pro iisdem valoribus cum y conspirans, $Y'-Y$ pro iisdem evanescere, adeoque per factores t, $t-\frac{1}{n}$, $t-\frac{2}{n}$, $t-\frac{3}{n}$, …, $t-1$ et proin etiam per eorum productum (quod est ordinis $n+1$) divisibilem esse, unde patet, Y', nisi prorsus identica sit cum Y, certo ad altiorem ordinem ascendere debere, sive Y ex omnibus functionibus integris ordinem n haud egredientibus unicam esse, quae pro illis $n+1$ valoribus cum y conspiret. Quodsi itaque y, in seriem secundum potestates ipsius t progredientem evoluta, ante terminum qui implicat t^{n+1} omnino abrumpitur, cum Y identica erit: si vero saltem tam cito convergit, ut terminos sequentes spernere liceat, functio Y inter limites $t=0$, $t=1$ sive $x=g$, $x=h$ ipsius y vice fungi poterit.

3.

Iam integrale nostrum $\int y\,dx$ transit in $L\int y\,dt$ a $t=0$ usque ad $t=1$ sumendum, cuius loco per ea, quae modo monuimus, adoptabimus $L\int Y\,dt$. Evolvendo itaque $Y^{(\mu)}$ in

$$\alpha t^n+\beta t^{n-1}+\gamma t^{n-2}+\delta t^{n-3}+\text{etc.}$$

erit $\int Y^{(\mu)}\,dt$, a $t=0$ usque ad $t=1$,

$$= M^{(\mu)}R^{(\mu)} = \frac{\alpha}{n+1} + \frac{\beta}{n} + \frac{\gamma}{n-1} + \frac{\delta}{n-2} + \text{etc.}$$

qua quantitate posita $= M^{(\mu)}R^{(\mu)}$, erit integrale quaesitum

$$= L(AR + A'R' + A''R'' + A'''R''' + \text{etc.} + A^{(n)}R^{(n)})$$

Exempli caussa apponemus computum coefficientis R'' pro $n=5$. Fit hic

$$T'' = 5t^5 - 13\cdot 5t^4 + 59\cdot 5t^3 - 107\cdot 5tt + 60\cdot 5t,$$
$$M'' = 2\cdot 1\cdot(-1)\cdot(-2)\cdot(-3) = -12$$

Hinc $-12R'' = \frac{5}{6} - \frac{1625}{5} + \frac{295}{4} - \frac{535}{3} + 150 = -\frac{5}{12}$, adeoque $R'' = \frac{5}{144}$.

Computus aliquando brevior evadit, statuendo $nt-1=u$. Tunc fit

$$T^{(\mu)} = \frac{(nu+n)(nu+n-1)(nu+n-2)\cdot\ldots\cdot(nu-n+4)(nu-n+3)(nu-n+2)(nu-n+1)}{2^n(nu+n-2\mu)}$$

Ponamus

$$\frac{(nnuu-nn)(nnuu-(n-2)^2)(nnuu-(n-4)^2)\cdot\ldots\cdot T\mu}{nnuu-(n-2\mu)^2}$$

ubi numerator desinere debet in $\ldots(nnuu-9)(nnuu-1)$, si n est impar, vel in $\ldots(nnuu-4)\cdot nu$, si n est par, eritque

$$M^{(\mu)} = \pm\frac{2^{n-1}(n-1)(n-2)\cdot\ldots\cdot 2\cdot 1}{(n-2\mu)}$$

Iam integrale $\int T^{(\mu)}\,dt$ a $t=0$ usque ad $t=1$ acceptum aequale est integrali $\frac{1}{2n}\int U^{(\mu)}\,du$ ab $u=-1$ usque ad $u=+1$.

Statuendo itaque

$$U^{(\mu)} = \alpha u^{n-1} + \beta u^{n-3} + \gamma u^{n-5} + \delta u^{n-7} + \text{etc.}$$

(sponte enim patet, potestates u^{n-2}, u^{n-4}, u^{n-6} etc. abesse), integralis pars $\int\alpha u^{n+1}\,du$ evanescet pro valore impari ipsius n, pars altera $\int\beta u^{n-1}\,du$ vero pro valore pari, unde integrale $\int T^{(\mu)}\,dt$ fiet pro n pari

$$\frac{2}{n}\Big(\frac{\alpha}{n+1} + \frac{\beta}{n-1} + \frac{\gamma}{n-3} + \frac{\delta}{n-5} + \text{etc.}\Big),$$

pro n impari autem

$$= \frac{2}{n}\Big(\frac{\alpha}{n+1} + \frac{\beta}{n-1} - \frac{\gamma}{n-3} - \text{etc.}\Big)$$

In exemplo nostro habetur

$$U'' = (25uu-25)(25uu-9) = 625u^4 - 850uu + 225,\ \text{adeoque}$$

$$-12R'' = -\frac{1}{12}(125 - 850 + 225) = \frac{5}{12}\ \text{ut supra.}$$

Observare convenit, fieri $U^{(n-\mu)} = \pm U^{(\mu)}$, adeoque $\int T^{(n-\mu)}\,dt = \pm \int T^{(\mu)}\,dt$, signo superiore valente pro n pari, inferiore pro impari. Quare quum facile perspiciatur, perinde haberi $M^{(n-\mu)} = \pm M^{(\mu)}$, semper erit $R^{(n-\mu)} = R^{(\mu)}$, sive e coefficientibus R, R', R'', ..., $R^{(n)}$ ultimus primo aequalis, penultimus secundo et sic porro.

4.

Valores numericos horum coefficientium a Cotesio usque ad $n = 10$ computatos ex Harmonia Mensurarum huc adscribimus.

Pro $n = 1$ sive terminis duobus.

$$R = R' = \tfrac{1}{2}$$

Pro $n = 2$ sive terminis tribus.

$$R = R'' = \tfrac{1}{6}, \quad R' = \tfrac{4}{6}$$

Pro $n = 3$ sive terminis quatuor.

$$R = R''' = \tfrac{1}{8}, \quad R' = R'' = \tfrac{3}{8}$$

Pro $n = 4$ sive terminis quinque.

$$R = R^{IV} = \tfrac{7}{90}, \quad R' = R''' = \tfrac{32}{90}, \quad R'' = \tfrac{12}{90}$$

Pro $n = 5$ sive terminis sex.

$$R = R^{V} = \tfrac{19}{288}, \quad R' = R^{IV} = \tfrac{75}{288}, \quad R'' = R''' = \tfrac{50}{288}$$

Pro $n = 6$ sive terminis septem.

$$R = R^{VI} = \tfrac{41}{840}, \quad R' = R^{V} = \tfrac{216}{840}, \quad R'' = R^{IV} = \tfrac{27}{840}, \quad R''' = \tfrac{272}{840}$$

Pro $n = 7$ sive terminis octo.

$$R = R^{VII} = \tfrac{751}{17280}, \quad R' = R^{VI} = \tfrac{3577}{17280}, \quad R'' = R^{V} = \tfrac{1323}{17280}, \quad R''' = R^{IV} = \tfrac{2989}{17280}$$

Pro $n = 8$ sive terminis novem.

$$R = R^{VIII} = \tfrac{989}{28350}, \quad R' = R^{VII} = \tfrac{5888}{28350}, \quad R'' = R^{VI} = -\tfrac{928}{28350}, \quad R''' = R^{V} = \tfrac{10496}{28350}, \quad R^{IV} = -\tfrac{4540}{28350}$$

Pro $n = 9$ sive terminis decem.

$$R = R^{IX} = \tfrac{2857}{89600}, \quad R' = R^{VIII} = \tfrac{15741}{89600}, \quad R'' = R^{VII} = \tfrac{1080}{89600}, \quad R''' = R^{VI} = \tfrac{19344}{89600}, \quad R^{IV} = R^{V} = \tfrac{5778}{89600}$$

Pro $n = 10$ sive terminis undecim.

$$R = R^{X} = \tfrac{16067}{598752}, \quad R' = R^{IX} = \tfrac{106300}{598752}, \quad R'' = R^{VIII} = -\tfrac{48525}{598752}, \quad R''' = R^{VII} = \tfrac{272400}{598752}, \quad R^{IV} = R^{VI} = -\tfrac{26055}{598752}, \quad R^{V} = \tfrac{427368}{598752}$$

5.

Quum formula $L(AR + A'R' + A''R'' + A'''R'''$ etc. $+A^{(n)}R^{(n)})$ integrale $\int y\,dx$ ab $x = g$ usque ad $x = h$, sive integrale $L\int y\,dt$ a $t = 0$ usque ad $t = 1$ exacte quidem exhibeat, quoties y in seriem evoluta potestatem t^n non transscendit, sed approxim ate tantum, quoties y ultra progreditur, superest, ut errorem, quem inducunt termini proxime sequentes, assignare doceamus.

Designemus generaliter per $k^{(m)}$ differentiam inter valorem verum integralis $\int t^m\,dt$ a $t = 0$ usque ad $t = 1$, atque valorem ex formula prodeuntem, ita ut sit

$$\begin{aligned}
k &= 1 - R - R' - R'' - R''' - \text{etc.} - R^{(n)} \\
k' &= \tfrac{1}{2} - \tfrac{1}{n}(R' + 2R'' + 3R''' + \text{etc.} + nR^{(n)}) \\
k'' &= \tfrac{1}{3} - \tfrac{1}{n^2}(R' + 4R'' + 9R''' + \text{etc.} + nnR^{(n)}) \\
k''' &= \tfrac{1}{4} - \tfrac{1}{n^3}(R' + 8R'' + 27R''' + \text{etc.} + n^3R^{(n)}) \\
&\text{etc.}
\end{aligned}$$

Patet igitur, si y evolvatur in seriem

$$K + K't + K''tt + K'''t^3 + \text{etc.}$$

differentiam inter valorem verum integralis $\int y\,dt$ atque valorem approximatum formulae exprimi per

$$Kk + K'k' + K''k'' + K'''k''' + \text{etc.}$$

Sed manifesto k, k', k'' etc. usque ad $k^{(n)}$ sponte fiunt $= 0$: correctio itaque formulae approximatae erit

$$K^{(n+1)}k^{(n+1)} + K^{(n+2)}k^{(n+2)} + K^{(n+3)}k^{(n+3)} + \text{etc.}$$

Indolem quantitatum $k^{(n+1)}$, $k^{(n+2)}$ etc. infra accuratius perscrutabimur; hic sufficiat, valores numericos primae aut secundae, pro singulis valoribus ipsius n, apposuisse, ut gradus praecisionis, quam formula approximata affert, inde aestimari possit.

Pro $n = 1$ habemus $k'' = -\frac{1}{12}$, $k''' = -\frac{1}{24}$, $k^{IV} = -\frac{19}{720}$
Pro $n = 2$ invenimus $k''' = 0$, $k^{IV} = -\frac{1}{180}$, $k^{V} = -\frac{1}{120}$
Pro $n = 3$ fit $k^{IV} = -\frac{3}{160}$, $k^{V} = -\frac{1}{80}$
Pro $n = 4 \ldots k^{V} = 0$, $k^{VI} = -\frac{8}{945}$, $k^{VII} = -\frac{1}{210}$
Pro $n = 5 \ldots k^{VI} = -\frac{275}{12096}$, $k^{VII} = -\frac{11}{1680}$
Pro $n = 6 \ldots k^{VII} = 0$, $k^{VIII} = -\frac{191}{12096}$, $k^{IX} = -\frac{1}{336}$
Pro $n = 7 \ldots k^{VIII} = -\frac{8183}{259200}$, $k^{IX} = -\frac{11}{3150}$
Pro $n = 8 \ldots k^{IX} = 0$, $k^{X} = -\frac{1472}{467775}$, $k^{XI} = -\frac{8}{6930}$
Pro $n = 9 \ldots k^{X} = -\frac{1}{44800}$, $k^{XI} = -\frac{8}{6300}$
Pro $n = 10 \ldots k^{XI} = 0$, $k^{XII} = -\frac{162147}{6918912}$, $k^{XIII} = -\frac{6912}{1201200}$

Pro valore pari ipsius n ubique hic fieri animadvertimus $k^{(n+1)} = 0$, ac praeterea $k^{(n+3)} = -\frac{n+3}{2}k^{(n+2)}$; pro valore impari ipsius n autem ubique prodit $k^{(n+2)} = -\frac{n+2}{2}k^{(n+1)}$. Ratio horum eventuum facile e considerationibus sequentibus depromitur.

Designemus generaliter per $l^{(m)}$ differentiam inter valorem verum huius integralis $\int(t - \frac{1}{2})^m\,dt$ a $t = 0$ usque ad $t = 1$, atque valorem eum, quem formula approximata profert, ita ut habeatur

$$l^{(m)} = \int(t - \tfrac{1}{2})^m\,dt - \Big((-\tfrac{1}{2})^m R + (\tfrac{1}{2n} - \tfrac{1}{2})^m R' + (\tfrac{2}{2n} - \tfrac{1}{2})^m R'' + (\tfrac{3}{2n} - \tfrac{1}{2})^m R''' + \text{etc.}\Big)$$

integrali a $t=0$ usque ad $t=1$ accepto. Manifesto pro valore impari ipsius m evanescet tum valor verus integralis tum valor approximatus: erit itaque $l'=0$, $l'''=0$, $l^V=0$, $l^{VII}=0$ etc. sive generaliter $l^{(m)}=0$ pro valore impari ipsius m. Pro valore pari autem m, formulae tribuimus formam hancce

$$l^{(m)} = \tfrac{1}{m+1}\cdot\tfrac{1}{2^{m+1}} - \Big((\tfrac{n}{2n})^m R + (\tfrac{n-2}{2n})^m R' + (\tfrac{n-4}{2n})^m R'' + \text{etc.}\Big)$$

si simul fuerit n pari; vel hanc

$$l^{(m)} = \pm\Big(\tfrac{1}{2^{m+1}} - (\tfrac{n-1}{2n})^m R - (\tfrac{n-3}{2n})^m R' - (\tfrac{n-5}{2n})^m R'' - \text{etc.}\Big)$$

si simul fuerit n impar.

Si igitur per evolutionem ipsius y in seriem secundum potestates ipsius $t-\frac{1}{2}$ progredientem prodit

$$y = L + L'(t-\tfrac{1}{2}) + L''(t-\tfrac{1}{2})^2 + L'''(t-\tfrac{1}{2})^3 + \text{etc.}$$

correctio valori approximato integralis $\int y\,dt$, a $t=0$ usque ad $t=1$ applicanda erit

$$Ll + L'l' + L''l'' + L'''l''' + \text{etc.}$$

aut potius, quum $l^{(m)}$ necessario evanescat pro valore quovis integro ipsius m haud maiore quam n, correctio erit

$$L^{(n+1)}l^{(n+1)} + L^{(n+2)}l^{(n+2)} + L^{(n+3)}l^{(n+3)} + \text{etc.}$$

pro n pari, vel

$$L^{(n+2)}l^{(n+2)} + L^{(n+3)}l^{(n+3)} + L^{(n+4)}l^{(n+4)} + \text{etc.}$$

pro n impari.

Facillime iam correctiones $l^{(m)}$ ad $k^{(m)}$ reducuntur et vice versa. Quum enim habeatur

$$(t-\tfrac{1}{2})^m = t^m - \frac{m}{1}\cdot\tfrac{1}{2}t^{m-1} + \frac{m(m-1)}{1\cdot 2}\cdot\tfrac{1}{4}t^{m-2} - \text{etc.}$$

erit

$$l^{(m)} = k^{(m)} - \frac{m}{2}k^{(m-1)} + \frac{m(m-1)}{8}k^{(m-2)} - \frac{m(m-1)(m-2)}{48}k^{(m-3)} + \text{etc.}$$

Et perinde fit

$$k^{(m)} = l^{(m)} + \frac{m}{2}l^{(m-1)} + \frac{m(m-1)}{8}l^{(m-2)} + \frac{m(m-1)(m-2)}{48}l^{(m-3)} + \text{etc.}$$

Ex posteriori formula eiicientur termini, ubi l afficitur indice impari: utraque autem continuanda est tantummodo usque ad indicem $n+1$ (inclus.). Manifesto itaque habebimus

pro n pari

$$k^{(n+1)} = l^{(n+1)}$$

$$k^{(n+2)} = l^{(n+2)} + \frac{n+2}{2}l^{(n+1)}$$

$$k^{(n+3)} = l^{(n+3)} + \frac{n+3}{2}l^{(n+2)} + \frac{(n+3)(n+2)}{8}l^{(n+1)}$$

pro n impari

$$k^{(n+1)} = l^{(n+1)}$$

$$k^{(n+2)} = l^{(n+2)}$$

$$k^{(n+3)} = l^{(n+3)} + \frac{n+3}{2}l^{(n+2)}$$

unde demanant observationes supra indicatae.

6.

Generaliter itaque loquendo praestabit, in applicanda methodo Cotesiana ipsi n tribuere valorem parem, seu terminorum multitudinem imparem in usum vocare. Perparum scilicet praecisio augebitur, si loco valoris paris ipsius n ad imparem proxime maiorem ascendamus, quum error maneat eiusdem ordinis, licet coefficiente aliquantulum minori affectus. Contra ascendendo a valore impari ipsius n ad parem proxime sequentem, error duobus ordinibus promovebitur, insuperque coefficiens notabilius imminutus praecisionem augebit. Ita si quinque termini adhibentur, sive pro $n = 4$, error proxime exprimitur per $-\frac{8}{945}L^{(5)}$ vel per $-\frac{2}{945}L^{(5)}$; si statuitur $n = 5$, error erit proxime $-\frac{275}{12096}L^{(6)}$ vel $= -\frac{55}{12096}L^{(6)}$, adeoque ne ad semissem quidem prioris depressus: contra faciendo $n = 6$, error fit proxime $= \frac{1}{140140}L^{(7)}$ vel $= -\frac{1}{210210}L^{(7)}$, praecisioque tanto magis aucta, quo citius series, in quam functio evolvitur, iam per se convergit.

7.

Postquam haecce circa methodum Cotesii praemisimus, ad disquisitionem generalem progredimur, abiiciendo conditionem, ut valores ipsius x progressione arithmetica procedant. Problema itaque aggredimur, determinare valorem integralis $\int y\,dx$ inter limites datos ex aliquot valoribus datis ipsius y, vel exacte vel quam proxime. Supponamus, integrale sumendum esse ab $x = g$ usque ad $x = h$, introducamusque loco ipsius x aliam variabilem $t = \frac{x-g}{h-g}$, ita ut integrale $L\int y\,dt$ a $t = 0$ usque ad $t = 1$ investigare oporteat. Respondeant $n+1$ valores dati ipsius y hi A, A', A'', A''', …, $A^{(n)}$ valoribus ipsius t inaequalibus hi a, a', a'', a''', …, $a^{(n)}$, designemusque per Y functionem algebraicam integram ordinis n hancce:

$$\begin{aligned}
Y = A \cdot \frac{(t-a')(t-a'')(t-a''')\cdot\ldots\cdot(t-a^{(n)})}{(a-a')(a-a'')(a-a''')\cdot\ldots\cdot(a-a^{(n)})} \\
+ A' \cdot \frac{(t-a)(t-a'')(t-a''')\cdot\ldots\cdot(t-a^{(n)})}{(a'-a)(a'-a'')(a'-a''')\cdot\ldots\cdot(a'-a^{(n)})} \\
+ A'' \cdot \frac{(t-a)(t-a')(t-a''')\cdot\ldots\cdot(t-a^{(n)})}{(a''-a)(a''-a')(a''-a''')\cdot\ldots\cdot(a''-a^{(n)})} \\
+ \text{etc.} \\
+ A^{(n)} \cdot \frac{(t-a)(t-a')(t-a'')\cdot\ldots\cdot(t-a^{(n-1)})}{(a^{(n)}-a)(a^{(n)}-a')(a^{(n)}-a'')\cdot\ldots\cdot(a^{(n)}-a^{(n-1)})}
\end{aligned}$$

Manifeste valores huius functionis, si t alicui quantitatum a, a', a'', a''', …, $a^{(n)}$ aequalis ponitur, coincidunt cum valoribus respondentibus functionis y, unde perinde ut in art. 2. concludimus, Y cum y identicam esse, quoties y quoque sit functio algebraica integra ordinem n non transscendens, aut saltem ipsius y vice fungi posse, si y in seriem secundum potestates ipsius t progredientem conversa tantam convergentiam exhibeat, ut terminos altiorum ordinum negligere liceat.

8.

Iam ad eruendum integrale $\int Y\,dt$ singulas partes ipsius Y consideremus. Designemus productum

$$(t-a)(t-a')(t-a'')(t-a''')\cdot\ldots\cdot(t-a^{(n)})$$

per T, fiatque per evolutionem huius producti

$$T = t^{n+1} + \alpha t^n + \alpha' t^{n-1} + \alpha'' t^{n-2} + \text{etc.} + \alpha^{(n)}$$

Numerator fractionis, per quam, in parte prima ipsius Y, multiplicata est A, fit $= \frac{T}{t-a}$; numeratores in partibus sequentibus perinde sunt $\frac{T}{t-a'}$, $\frac{T}{t-a''}$, $\frac{T}{t-a'''}$ etc. Denominatores vero nihil aliud sunt, nisi valores determinati horum numeratorum, si resp. statuitur $t = a$, $t = a'$, $t = a''$, $t = a'''$ etc.: denotemus hos denominatores resp. per M, M', M'', M''' etc., ita ut sit

$$Y = \frac{A}{M} \cdot \frac{T}{t-a} + \frac{A'}{M'} \cdot \frac{T}{t-a'} + \frac{A''}{M''} \cdot \frac{T}{t-a''} + \frac{A'''}{M'''} \cdot \frac{T}{t-a'''} + \text{etc.} + \frac{A^{(n)}}{M^{(n)}} \cdot \frac{T}{t-a^{(n)}}$$

Quum fiat $T = 0$, pro $t = a$, habemus aequationem identicam

$$\alpha^{(n)} + \alpha a^n + \alpha' a^{n-1} + \alpha'' a^{n-2} + \text{etc.} + \alpha^{(n)} = 0$$

adeoque

$$\begin{aligned} T = t^{n+1} - a^{n+1} + \alpha(t^n - a^n) + \alpha'(t^{n-1} - a^{n-1}) + \alpha''(t^{n-2} - a^{n-2}) + \text{etc.} \\ + \alpha^{(n-1)}(t-a) \end{aligned}$$

Hinc dividendo per $t - a$ fit

$$\begin{aligned} \frac{T}{t-a} = t^n + a t^{n-1} + a a' t^{n-2} + a a'' t^{n-3} + \text{etc.} + a a^{(n-1)} \\ + \alpha' t^{n-1} + \alpha' a' t^{n-2} + \alpha' a'' t^{n-3} + \text{etc.} + \alpha' a^{(n-2)} \\ + \alpha'' t^{n-2} + \alpha'' a'' t^{n-3} + \text{etc.} + \alpha'' a^{(n-3)} \\ + \text{etc.} \end{aligned}$$

Valor huius functionis pro $t = a$ colligitur

$$\begin{aligned} = (n+1)a^n + (n\alpha + (n-1)\alpha')a^{n-1} + ((n-1)\alpha' + (n-2)\alpha'')a^{n-2} + \text{etc.} \\ + \alpha^{(n-1)} \end{aligned}$$

Hinc M aequalis valori ipsius $\frac{dT}{dt}$ pro $t = a$, uti etiam aliunde constat. Perinde M', M'', M''' etc. erunt valores ipsius $\frac{dT}{dt}$ pro $t = a'$, $t = a''$, $t = a'''$ etc.

Porro invenimus valorem integralis $\int \frac{T\,dt}{t-a}$, a $t = 0$ usque ad $t = 1$,

$$\begin{aligned} = \frac{1}{n+1} + \frac{a}{n} + \frac{aa'}{n-1} + \frac{aa''}{n-2} + \text{etc.} + aa^{(n-1)} \\ + \frac{1}{n}\alpha' + \frac{\alpha' a'}{n-1} + \frac{\alpha' a''}{n-2} + \text{etc.} + \alpha' a^{(n-2)} \\ + \frac{1}{n-1}\alpha'' + \frac{\alpha'' a''}{n-2} + \text{etc.} + \alpha'' a^{(n-3)} \\ + \text{etc.} \end{aligned}$$

os terminos ordine sequenti disponentes:

$$\frac{a^n + \alpha a^{n-1} + \alpha' a^{n-2} + \alpha'' a^{n-3} + \text{etc.} + \alpha^{(n-1)}}{n+1}$$
$$+ \frac{1}{n}(a^{n-1} + \alpha a^{n-2} + \alpha' a^{n-3} + \text{etc.} + \alpha^{(n-2)})$$
$$+ \frac{1}{n-1}(a^{n-2} + \alpha a^{n-3} + \alpha' a^{n-4} + \text{etc.} + \alpha^{(n-3)})$$
$$+ \frac{1}{n-2}(a^{n-3} + \alpha a^{n-4} + \alpha' a^{n-5} + \text{etc.} + \alpha^{(n-4)})$$
$$+ \text{etc.}$$
$$+ \frac{1}{2}(\alpha^{(n-2)} + \alpha^{(n-1)})$$

Sed manifesto eadem quantitas prodit, si in producto e multiplicatione functionis T in seriem infinitam

$$t^{-1} + \tfrac{1}{2}t^{-2} + \tfrac{1}{3}t^{-3} + \tfrac{1}{4}t^{-4} + \text{etc.}$$

orto, reiectis omnibus terminis, qui implicant potestates ipsius t exponentibus negativis (sive brevius, in producti parte ea, quae est functio integra ipsius t) pro t scribitur a. Supponamus itaque, fieri*)

$$T(t^{-1} + \tfrac{1}{2}t^{-2} + \tfrac{1}{3}t^{-3} + \tfrac{1}{4}t^{-4} + \text{etc.}) = T' + T''$$

ita ut T' sit functio integra ipsius t in hoc producto contenta, T'' vero pars altera, scilicet series descendens in infinitumque excurrens. Quo facto valor integralis $\int \frac{T\,dt}{t-a}$ a $t = 0$ usque ad $t = 1$ aequalis erit valori functionis T'' pro $t = a$. Quodsi itaque valores determinatos functionis

$$\frac{T''}{\frac{dT}{dt}}$$

pro $t = a$, $t = a'$, $t = a''$, $t = a'''$ etc. usque ad $t = a^{(n)}$ resp. per R, R', R'', R''', $\ldots$, $R^{(n)}$ denotamus, integrale $\int Y\,dt$ a $t = 0$ usque ad $t = 1$ fiet

$$= AR + A'R' + A''R'' + A'''R''' + \text{etc.} + A^{(n)}R^{(n)}$$

quod per L multiplicatum exhibebit valorem verum vel approximatum integralis $\int y\,dx$ ab $x = g$ usque ad $x = h$.

*) Vix opus erit monere, characteres T, T', T'' alio sensu hic accipi, quam in art. 2.

9.

Hae operationes aliquanto facilius perficiuntur, si loco indeterminatae t introducitur alia $u = 2t - 1$. Scribimus quoque brevitatis caussa $b = 2a - 1$, $b' = 2a' - 1$, $b'' = 2a'' - 1$ etc.

Transeat T in U Substitute pro t valore $\frac{u+1}{2}$, in $2^{-n-1}U$, sive sit

$$U = (u - b)(u - b')(u - b'')(u - b''') \cdot \ldots \cdot (u - b^{(n)})$$

Erit itaque $\frac{dt}{t-a} = \frac{du}{u-b}$, atque M, M', M'' etc. valores determinati ipsius $\frac{dU}{du}$, si deinceps statuitur $u = b$, $u = b'$, $u = b''$ etc.

Quum series $t^{-1}+\frac{1}{2}t^{-2}+\frac{1}{3}t^{-3}+\frac{1}{4}t^{-4}+$etc. nihil aliud sit quam $\frac{1}{1-t}=-\frac{1}{t}\log(1-\frac{1}{t})$, per substitutionem $t=\frac{u+1}{2}$ necessario transibit in $2u^{-1}+\frac{2}{2}u^{-2}+\frac{2}{3}u^{-3}+\frac{2}{4}u^{-4}+$etc.. Quodsi itaque statuimus

$$U(u^{-1}+\tfrac{1}{2}u^{-2}+\tfrac{1}{3}u^{-3}+\tfrac{1}{4}u^{-4}+\text{etc.})=U'+U''$$

ita ut U' sit functio integra ipsius u in hoc producto contenta, U'' vero pars altera, puta series descendens infinita, patet esse

$$\frac{U''}{\frac{dU}{du}}=\frac{2^nT''}{\frac{dT}{dt}}$$

Sed manifesto T', tamquam functio integra ipsius t, per substitutionem $t=\frac{u+1}{2}$ necessario functionem integram ipsius u producet: contra T'', quae non continet nisi potestates negativas ipsius t, per eandem substitutionem tantummodo potestates negativas ipsius u gignet. Quam ob rem U' nihil aliud erit quam $2^nT'$ per hanc substitutionem transformata, ac perinde U'' producta erit ex $2^nT''$. Nihil itaque intererit, sive in $\frac{T''}{\frac{dT}{dt}}$ substituamus $t=a$, sive in $\frac{U''}{\frac{dU}{du}}$ faciamus $u=b$, unde colligimus, R, R', R'', R''' etc. etiam esse valores determinatos functionis $\frac{U''}{\frac{dU}{du}}$ pro $u=b$, $u=b'$, $u=b''$, $u=b'''$ etc.

10.

Antequam ulterius progrediamur, haecce praecepta per exemplum illustrabimus. Sit $n=5$, statuamusque $a=0$, $a'=\frac{1}{5}$, $a''=\frac{2}{5}$, $a'''=\frac{3}{5}$, $a^{IV}=\frac{4}{5}$, $a^{V}=1$.

Hinc fit

$$\begin{aligned}T&=t(t-\tfrac{1}{5})(t-\tfrac{2}{5})(t-\tfrac{3}{5})(t-\tfrac{4}{5})(t-1)\\&=t^6-3t^5+\tfrac{12}{5}t^4-\tfrac{12}{5}t^3+\tfrac{47}{25}t^2-\tfrac{12}{25}t\end{aligned}$$

Multiplicando per $t^{-1}+\frac{1}{2}t^{-2}+\frac{1}{3}t^{-3}+\frac{1}{4}t^{-4}+\frac{1}{5}t^{-5}+\frac{1}{6}t^{-6}+$etc. obtinemus

$$\begin{aligned}T'&=-\tfrac{1}{6}t^5+\tfrac{5}{12}t^4-\tfrac{137}{180}t^3+\tfrac{15}{16}t^2-\tfrac{2003}{3600}t+\tfrac{137}{1800}\\T''&=\tfrac{1}{6}t^{-1}-\tfrac{5}{4}t^{-2}+\tfrac{137}{36}t^{-3}-15t^{-4}+\tfrac{4006}{75}t^{-5}-\tfrac{548}{5}t^{-6}+\text{etc.}\end{aligned}$$

Valores itaque coefficientium R, R', R'', R''', R^{IV}, R^{V} exprimuntur per functionem fractam

$$\frac{\frac{1}{6}t^{-1}-\frac{5}{4}t^{-2}+\frac{137}{36}t^{-3}-15t^{-4}+\frac{4006}{75}t^{-5}-\frac{548}{5}t^{-6}+\text{etc.}}{6t^5-15t^4+\frac{48}{5}t^3-\frac{36}{5}t^2+\frac{94}{25}t-\frac{12}{25}}$$

in qua pro t deinceps substituendi sunt valores 0, $\frac{1}{5}$, $\frac{2}{5}$, $\frac{3}{5}$, $\frac{4}{5}$, 1. Aliquanto brevior est methodus altera, quae suppeditat $b=-1$, $b'=-\frac{3}{5}$, $b''=-\frac{1}{5}$, $b'''=+\frac{1}{5}$, $b^{IV}=+\frac{3}{5}$, $b^{V}=+1$,

$$U=(uu-1)(uu-\tfrac{9}{25})(uu-\tfrac{1}{25})=u^6-\tfrac{7}{5}u^4+\tfrac{259}{625}u^2-\tfrac{9}{625}$$

unde R, R', R'' etc. erunt valores functionis fractae

$$\frac{\frac{1}{6}u^{-1}-\frac{7}{30}u^{-3}+\frac{259}{1875}u^{-5}-\frac{9}{625}u^{-7}+\text{etc.}}{6u^5-\frac{28}{5}u^3+\frac{518}{625}u}$$

pro $u=-1$, $u=-\frac{3}{5}$, $u=-\frac{1}{5}$ etc. Utraque methodus eosdem numeros profert, quos in art. 4. ex Harmonia Mensurarum tradidimus. Ceterum in casu tali, qualem hocce exemplum sistit, ubi a, a', a'' etc. sunt quantitates rationales, valores denominatoris M commodius in forma primitiva computantur, puta $(a-a')(a-a'')(a-a''')\cdot\ldots\cdot(a-a^{(n)})$ pro $t=a$ ac perinde de reliquis. Idem valet de denominatore $\frac{dU}{du}$, qui pro $u=b$ fit $=(b-b')(b-b'')(b-b''')\cdot\ldots\cdot(b-b^{(n)})$.

11.

Quoties a, a', a'' etc. vel ex parte vel omnes sunt irrationales, utilis erit transformatio functionis fractae, ex qua numeros R, R', R'' etc. derivamus, in functionem integram: principia talis transformationis, quum in libris algebraicis non inveniantur, hoc loco breviter explicabimus. Propositis scilicet tribus functionibus integris Z, C, C' indeterminatae z, quaeritur functio integra, quae fractae $\frac{Z}{C'}$ vice fungi possit, quatenus pro z accipitur radix quaecunque aequationis $C=0$. Supponemus autem, C pro nullo horum valorum ipsius z evanescere, sive quod eodem redit, C atque C' nullum divisorem communem indeterminatum implicare. Exponentes potestatum altissimarum ipsius z in C atque C' per k, k' denotabimus.

Dividatur sueto more C per C', donec residui ordo infra k' depressus sit; statuatur residuum $=\gamma C''$, eiusque ordo $=k''$, ita ut $\gamma z^{k''}$ sit residui terminus altissimus; divisionis quotientem ponemus $=p$. Perinde ex divisione functionis C' per C'' prodeat residuum $\gamma' C'''$ ordinis k''', quotiens $-p'$, tum rursus e divisione functionis C'' per C''' prodeat residuum $\gamma'' C^{IV}$ ordinis k^{IV} atque quotiens $-p''$ et sic porro, donec in serie functionum C'', C''', C^{IV}, C^{V} etc., quae singulae terminum suum altissimum coefficiente 1 affectum habebunt, perveniatur ad $C^{(m+1)}=1$. Hoc tandem evenire debere facile perspicitur, quum quaelibet functionum C, C', C'', C''' etc. cum praecedenti divisorem communem indeterminatum habere nequeat, adeoque certo divisio absque residuo fieri nequeat, quamdiu divisor fuerit ordinis maioris quam 0. Habebimus igitur seriem aequationum

$$\begin{gathered}
C=pC'-\gamma C''\\
C'=p'C''-\gamma' C'''\\
C''=p''C'''-\gamma'' C^{IV}\\
C'''=p'''C^{IV}-\gamma''' C^{V}\\
\text{etc. usque ad}\\
C^{(m-1)}=p^{(m-1)}C^{(m)}-\gamma^{(m-1)}C^{(m+1)}
\end{gathered}$$

ubi C'', C''', C^{IV}, C^{V}, $\ldots$, $C^{(m+1)}$ sunt functiones integrae ipsius z ordinis k'', k''', k^{IV}, k^{V}, $\ldots$, $k^{(m+1)}$; numeri k', k'', k''', $\ldots$, $k^{(m+1)}$ continuo decrescentes usque ad ultimum $k^{(m+1)}=0$; p, p', p'', p''' etc. quoque functiones integrae ipsius z ordinis $k-k'$, $k'-k''$, $k''-k'''$, $k'''-k^{IV}$ etc. (excepto casu, ubi $k<k'$, in quo manifeste statui debet $p=0$).

His ita praeparatis formamus secundam seriem functionum integrarum ipsius z, puta η, η', η'', η''', $\ldots$, $\eta^{(m+1)}$. Et quidem statuemus $\eta=1$, $\eta'=0$, reliquas vero singulas e binis praecedentibus per eandem legem derivamus, per quam functiones C, C', C'', C''' etc. inter se nexae sunt, scilicet per aequationes

$$\begin{gathered}
\eta''=p\eta'-\gamma\eta\\
\eta'''=p'\eta''-\gamma'\eta'\\
\eta^{IV}=p''\eta'''-\gamma''\eta''\\
\text{etc. usque ad}\\
\eta^{(m+1)}=p^{(m-1)}\eta^{(m)}-\gamma^{(m-1)}\eta^{(m-1)}
\end{gathered}$$

Manifesto $\eta^{(m+1)}$ hic est ordinis 0; $\eta^{(m)} = -p^{(m-1)}$ ordinis $k^{(m-1)}$, et perinde sequentes $\eta^{(m-1)}$, $\eta^{(m-2)}$ etc. resp. ordinis $k' - k'''$, $k' - k^{IV}$ etc., ita ut ultima η ascendat ad ordinem $k' - k^{(m)}$.

Porro consideremus tertiam functionum seriem, $C\eta - C'\eta'$, $C\eta' - C'\eta''$, $C\eta'' - C''\eta'''$, $C\eta''' - C'''\eta^{IV}$ etc., inter cuius terminos quosvis ternos consequentes manifesto similis relatio intercedet, scilicet

$$\begin{aligned}
C\eta'' - C''\eta &= p(C\eta' - C'\eta'') - \gamma(C\eta - C'\eta') \\
C\eta''' - C'''\eta' &= p'(C\eta' - C'\eta'') - \gamma'(C\eta'' - C''\eta''') \\
C\eta^{IV} - C^{IV}\eta'' &= p''(C\eta'' - C''\eta''') - \gamma''(C\eta''' - C'''\eta^{IV}) \\
&\text{etc.}
\end{aligned}$$

Iam prima harum functionum fit $= 0$, secunda $= C'$: hinc facile colligitur, singulas per C' divisibiles fore.

Hinc autem nullo negotio sequitur, loco fractionis $\frac{Z}{C'}$ adoptari posse functionem integram $Z\eta^{(m+1)}$, quatenus quidem ipsi z non tribuantur alii valores nisi qui sint radices aequationis $C' = 0$: manifesto enim differentia $\frac{Z}{C'} - Z\eta^{(m+1)}$ pro tali valore ipsius z necessario evanescit, quum $1 - C'\eta^{(m+1)} = C^{(m+1)} - C'\eta^{(m+1)}$ per C' sit divisibilis.

Loco functionis $Z\eta^{(m+1)}$ etiam adoptari poterit eius residuum ex divisione per C ortum, cuius ordo erit inferior ordine functionis C'.

Ceterum hocce residuum commodius per algorithmum sequentem immediate eruere licet. Formentur aequationes sequentes

$$\begin{aligned}
Z &= qC' + Z' \\
Z' &= q'C'' + Z'' \\
Z'' &= q''C''' + Z''' \\
Z''' &= q'''C^{IV} + Z^{IV} \text{ etc. usque ad}
\end{aligned}$$

scilicet deinceps dividendo Z per C', dein residuum primae divisionis Z' per C'', tum residuum secundae divisionis per C''' ac sic porro. Quum residuum semper ad ordinem inferiorem pertineat quam divisor, ordo functionum Z', Z'', Z''', Z^{IV} etc. erit resp. inferior quam k', k'', k''', k^{IV} etc.; ultima vero $Z^{(m+1)}$ necessario fit $= 0$, quum divisor $C^{(m+1)}$ sit $= 1$. Habemus itaque

$$Z = q'C' + q''C'' + q'''C''' + q^{IV}C^{IV} + \text{etc.} + q^{(m+1)}C^{(m+1)}$$

Quatenus autem pro z solae radices aequationis $C' = 0$ accipiuntur, fit $C' = 0$, $C'' = C'\eta''$, $C''' = C'\eta'''$, $C^{IV} = C'\eta^{IV}$ etc., unde sub eadem restrictione erit

$$\frac{Z}{C'} = q'\eta' + q''\eta'' + q'''\eta''' + q^{IV}\eta^{IV} + \text{etc.} + q^{(m+1)}\eta^{(m+1)}$$

Ordo vero huius expressionis necessario erit infra k': quum enim ordo quotientium q'', q''', q^{IV} etc. esse debeat infra $k' - k''$, $k'' - k'''$, $k''' - k^{IV}$ etc., ordo singularum partium $q''\eta''$, $q'''\eta'''$, $q^{IV}\eta^{IV}$ etc. erit infra $k' - k''$, $k' - k'''$, $k' - k^{IV}$ etc.

Denique adhuc observamus, si forte inter valores indeterminatae z, quos in fractione $\frac{Z}{C'}$ substituere oporteat, rationales cum irrationalibus mixti reperiantur, magis e re fore, illos ab his separare atque hos solos in aequatione $C' = 0$ comprehendere. Pro rationalibus enim valoribus calculi compendio opus non erit; pro irrationalibus autem calculus tanto simplicior erit, quo minor fuerit gradus functionis integrae, ad quam fractam reducere licet.

12.

Ecce nunc exemplum transformationis in art. praec. explicatae. Proposita sit functio fracta
$$\frac{z^3-8z^2+105zz-81z+22}{7z^4-10z^3+81zz-88z+88}$$
in qua z indefinite repraesentat radices aequationis $7z^4-10z^3+81zz-88z+88=0$.

Si hic omnes septem radices complecti vellemus, ad functionem integram sexti ordinis delaberemur. Manifesto autem pro valore rationali $z=0$ computus fractionis obvius est, datque valorem $\frac{22}{88}=\frac{1}{4}$: quapropter seposita hac radice in aequatione sexti gradis subsistemus:
$$\frac{Z}{C'}=\frac{z^2-8z+105z-\frac{81}{z}+\frac{22}{z^2}}{7z^4-10z^3+81zz-88z+88}$$
quo pacto facile praevidemus orituram esse functionem integram quarti ordinis.

Iam ex applicatione praeceptorum praecedentium prodeunt sequentia:
$$\begin{array}{rlr}
C &= z^4-\frac{10}{7}z^3+\frac{81}{7}zz-\frac{88}{7}z+\frac{88}{7} & k=4\\
C' &= z^3-8z^2+105z-81+\frac{22}{z} & k'=3\\
C'' &= zz-\frac{63}{41}z+\frac{1586}{451}-\frac{7128}{4961}\cdot\frac{1}{z}+\frac{10648}{54571}\cdot\frac{1}{zz} & k''=2\\
C''' &= z-\frac{2044}{451}+\frac{7128}{451}\cdot\frac{1}{z}-\frac{10648}{451}\cdot\frac{1}{zz} & k'''=1\\
C^{IV} &= 1 & k^{IV}=0
\end{array}$$

unde
$$\begin{array}{rlrl}
p &= \frac{1}{7}z+\frac{354}{451} & \gamma &= \frac{451}{7\cdot 41}\\
p' &= \frac{41}{7}z-\frac{203}{451} & \gamma' &= -\frac{451^2}{7^2\cdot 41}\\
p'' &= -\frac{7^2\cdot 41}{451}z+\frac{2044}{451} & \gamma'' &= -\frac{451^3}{7^3\cdot 41^2}
\end{array}$$

$$\begin{array}{rlr}
\eta &= 1, & \eta'=0,\\
\eta'' &= \frac{1}{7}z+\frac{354}{451}, &\\
\eta''' &= \frac{41}{49}zz-\frac{143}{451\cdot 7}z-\frac{451\cdot 203+354\cdot 41^2}{451^2}, &\\
\eta^{IV} &= -\frac{41}{343}z^3+\frac{6\cdot 451+3\cdot 41\cdot 203}{7^2\cdot 451}zz-\frac{3\cdot 451\cdot 203\cdot 354+41^2\cdot 10648}{7\cdot 451^3}z &\\
&\quad -\frac{451^2\cdot 203+354^2\cdot 41^2+451\cdot 41\cdot 10648}{451^4}. &
\end{array}$$

Hinc tandem derivatur functio integra fractioni propositae aequivalens:
$$\frac{1}{4}+\frac{451}{16}z-\frac{41\cdot 7}{64}zz+\frac{7^2\cdot 41}{256}z^3-\frac{7^3\cdot 41^2}{1024}z^4$$

13.

Ad determinandum gradum praecisionis, qua formula nostra integralis $L(AR+A'R'+A''R''+\text{ etc. }+A^{(n)}R^{(n)})$ gaudet, statuamus generaliter
$$R\alpha^n+R'\alpha'^n+R''\alpha''^n+\text{etc.}+R^{(n)}\alpha^{(n)n}=k^{(n)}$$

ita ut $k^{(m)}$ sit differentia inter integralis $\int t^m\,dt$ a $t=0$ usque ad $t=1$ sumti valorem verum atque approximatum. Habebimus itaque, singulis fractionibus in series evolutis,

$$\begin{aligned} T\Big(t^{-1}+\tfrac{1}{2}t^{-2}+\tfrac{1}{3}t^{-3}+\tfrac{1}{4}t^{-4}+\text{etc.}\Big)-T' &= k^{(0)}t^{-1}+k^{(1)}t^{-2}+k^{(2)}t^{-3}+k^{(3)}t^{-4}+\text{etc.}\\ &= (1-k)t^{-1}+(1-k')t^{-2}+(1-k'')t^{-3}+(1-k''')t^{-4}+\text{etc.}\\ &= t^{-1}+\tfrac{1}{2}t^{-2}+\tfrac{1}{3}t^{-3}+\tfrac{1}{4}t^{-4}+\text{etc.}-\Theta \end{aligned}$$

si statuamus

$$\Theta = kt^{-1}+k't^{-2}+k''t^{-3}+k'''t^{-4}+\text{etc.}$$

sive potius (quum iam sciamus, k, k', k'', k''' etc. usque ad $k^{(n)}$ sponte evanescere debere)

$$\Theta = k^{(n+1)}t^{-(n+2)}+k^{(n+2)}t^{-(n+3)}+k^{(n+3)}t^{-(n+4)}+\text{etc.}$$

Multiplicando per T fit

$$T\Theta = k^{(n+1)}Tt^{-(n+2)}+k^{(n+2)}Tt^{-(n+3)}+k^{(n+3)}Tt^{-(n+4)}+\text{etc.}$$

Pars prior huius aequationis est functio integra ipsius t ordinis n, eiusque valores determinati pro $t=a$, $t=a'$, $t=a''$ etc. resp. fiunt MR, $M'R'$, $M''R''$ etc.: quapropter, quum eadem valeant de functione T', uti ex ipso modo numeros R, R', R'' etc. determinandi perspicuum est, necessario illa pars prior aequationis identica esse debet cum T', adeoque $T''=T\Theta$. Oritur itaque Θ ex evolutione fractionis $\frac{T''}{T}$, quo pacto coefficientes $k^{(1)}$, $k^{(2)}$ etc. quousque libet determinari poterunt. Quibus inventis correctio valoris nostri approximati integralis $\int y\,dt$ erit

$$K^{(n+1)}k^{(n+1)}+K^{(n+2)}k^{(n+2)}+K^{(n+3)}k^{(n+3)}+\text{etc.}$$

si series, in quam evolvitur y, est

$$y = K+K't+K''tt+K'''t^3+\text{etc.}$$

14.

Si magis placet, correctionem exprimere per coefficientes seriei secundum potestates ipsius $t-\frac{1}{2}$ progredientis

$$y = L+L'(t-\tfrac{1}{2})+L''(t-\tfrac{1}{2})^2+L'''(t-\tfrac{1}{2})^3+\text{etc.}$$

illa erit

$$L^{(n+1)}l^{(n+1)}+L^{(n+2)}l^{(n+2)}+L^{(n+3)}l^{(n+3)}+\text{etc.}$$

si generaliter per $l^{(m)}$ exprimimus correctionem valoris approximati integralis $\int(t-\frac{1}{2})^m\,dt$. Hae correctiones $l^{(m)}$ cum correctionibus $k^{(m)}$ nexae erunt per aequationem

$$\begin{aligned} l^{(m)} = k^{(m)}-\frac{m}{2}k^{(m-1)}+\frac{m(m-1)}{2\cdot 4}k^{(m-2)}-\frac{m(m-1)(m-2)}{2\cdot 4\cdot 6}k^{(m-3)}+\text{etc.}\\ +(-1)^m\frac{m(m-1)(m-2)\cdot\ldots\cdot 2\cdot 1}{2\cdot 4\cdot 6\cdot\ldots\cdot 2m}k^{(0)}. \end{aligned}$$

Quo vero illas independenter eruere possimus, perpendamus, functionem Θ per substitutionem $t = \frac{u+1}{2}$ transire in

$$2k\,u^{-1} + 4(k' - \tfrac{1}{2}k)u^{-2} + 8(k'' - \tfrac{2}{2}k' + \tfrac{1\cdot 3}{2\cdot 4}k)u^{-3} \\ + 16(k''' - \tfrac{3}{2}k'' + \tfrac{3\cdot 5}{2\cdot 4}k' - \tfrac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}k)u^{-4} + \text{etc.}$$

sive in

$$2l\,u^{-1} + 4l'u^{-2} + 8l''u^{-3} + 16l'''u^{-4} + \text{etc.}$$

sive denique, quum a priori sciamus, l, l', l'', l''' etc. usque ad $l^{(n)}$ sponte evanescere, in

$$2^{n+2}l^{(n+1)}u^{-(n+2)} + 2^{n+3}l^{(n+2)}u^{-(n+3)} + 2^{n+4}l^{(n+3)}u^{-(n+4)} + \text{etc.}$$

At $T' = \frac{T''}{\Theta}$; quare quum T, T'' per substitutionem $t = \frac{u+1}{2}$ transeant in U, U'' (art. 9), functio Θ per eandem substitutionem transibit in $\frac{U''}{U}$. Quodsi itaque seriem ex evolutione fractionis $\frac{U''}{U}$ oriundam per Q designamus, erit

$$Q = 2^{n+1}l^{(n+1)}u^{-(n+2)} + 2^{n+2}l^{(n+2)}u^{-(n+3)} + 2^{n+3}l^{(n+3)}u^{-(n+4)} + \text{etc.}$$

quo pacto coefficientes $l^{(n+1)}$, $l^{(n+2)}$, $l^{(n+3)}$ etc. quousque lubet erui poterunt.

Ita in exemplo art. 10 invenimus

$$\begin{aligned} U'' &= -\tfrac{1}{6}u^{-1} + \tfrac{7}{30}u^{-3} - \tfrac{259}{1875}u^{-5} + \tfrac{9}{625}u^{-7} - \text{etc.} \\ Q &= -\tfrac{1}{6}u^{-1} + \tfrac{7}{30}u^{-3} - \tfrac{1058}{9375}u^{-5} + \tfrac{6933}{78125}u^{-7} - \text{etc.} \end{aligned}$$

adeoque correctio valoris approximati integralis

$$\tfrac{1}{15120}L^{(5)} - \tfrac{1}{112500}L^{(7)} + \tfrac{6933}{1367187500}L^{(9)} - \text{etc.}$$

15.

Coefficiens $K^{(m)}$ functionis y in seriem evolutae fit, per theorema Taylori, aequalis valori ipsius

$$\frac{1}{1\cdot 2\cdot 3\cdot \ldots \cdot m}\cdot\frac{d^m y}{dt^m} \quad \text{sive} \quad \frac{L^m}{1\cdot 2\cdot 3\cdot \ldots \cdot m}\cdot\frac{d^m y}{dx^m}$$

pro $t = 0$ sive $x = g$; perinde coefficiens $L^{(m)}$ est valor eiusdem expressionis pro $t = \frac{1}{2}$ sive $x = g + \frac{1}{2}L$: utrique coefficienti ordinem m tribuemus.

Generaliter itaque loquendi integratio nostra usque ad ordinem n inclus. exacta erit, quicunque valores pro a, a', a'', …, $a^{(n)}$ accipiantur. Attamen hinc nihil obstat, quominus pro valoribus harum quantitatum scite electis praecisio ad altiorem gradum evehatur. Ita iam supra vidimus, in methodo Cotesii i. e. pro $a = 0$, $a' = \frac{1}{n}$, $a'' = \frac{2}{n}$, $a''' = \frac{3}{n}$ etc. praecisionem sponte ad ordinem $n+1$ inclus. extendi, quoties n sit numerus par. Generaliter patet, si valores a, a', a'', a''' etc. ita fuerint electi, ut in functione T'' vel U'' ab initio excidat terminus unus pluresve, praecisionem totidem gradibus ultra ordinem n promotum iri, quot termini exciderint. Hinc facile colligitur, quum multitudo quantitatum quas eligere conceditur sit $n+1$, per idoneam earum determinationem praecisionem semper ad ordinem $2n+1$ inclus. evehi posse, quo pacto adiumento $n+1$ terminorum eundem praecisionis ordinem assequi licebit, ad quem attingendum $2n+1$ vel $2n+2$ terminos in usum vocare oporteret, si Cotesii methodum sequeremur.

16.

Totum hoc negotium in eo vertitur, ut pro quovis valore dato ipsius n functionem T eruamus formae $t^{n+1} + \alpha t^n + \alpha' t^{n-1} + \alpha'' t^{n-2} + \ldots$ ita comparatam, ut in producto

$$T(t^{-1} + \tfrac{1}{2}t^{-2} + \tfrac{1}{3}t^{-3} + \tfrac{1}{4}t^{-4} + \text{etc.})$$

evoluto potestates t^0, t^{-1}, t^{-2}, ..., $t^{-(2n+1)}$ omnes nanciscantur coefficientem 0; aut si magis placet, functionem U formae $u^{n+1} + \beta u^n + \beta' u^{n-1} + \beta'' u^{n-2} + \ldots$, cuius productum per $u^{-1} + \frac{1}{2}u^{-2} + \frac{1}{3}u^{-3} + \frac{1}{4}u^{-4}+$ etc. liberum evadat a potestatibus u^0, u^{-1}, u^{-2}, u^{-3}, ..., $u^{-(2n+1)}$. Modus posterior aliquanto simplicior erit: quum enim facile perspiciatur, coefficientes ipsius U, ut conditioni praescriptae satisfiat, alternatim evanescere debere, sive statui $\beta = 0$, $\beta'' = 0$, $\beta^{IV} = 0$ etc., laboris dimidia fere pars iam absoluta censenda erit. Evolvamus casus quosdam simpliciores.

I. Pro $n = 0$, coefficiens unicus ipsius t^{-1} in producto $(t+a)(t^{-1}+\frac{1}{2}t^{-2}+\frac{1}{3}t^{-3}+\frac{1}{4}t^{-4}+\text{etc.})$ evanescere debet. Qui quum fiat $= 1 + a$, habemus $a = -\frac{1}{2}$, sive $T = t - \frac{1}{2}$. Perinde $U = u$.

II. Pro $n = 1$, determinatio ipsius T pendet a duabus aequationibus

$$\begin{aligned} 0 &= 1 + \tfrac{1}{2}\alpha + \alpha' \\ 0 &= \tfrac{1}{2} + \tfrac{1}{3}\alpha + \tfrac{1}{2}\alpha' \end{aligned}$$

unde deducimus $\alpha = -1$, $\alpha' = +\frac{1}{6}$, sive $T = tt - t + \frac{1}{6}$. Determinatio functionis U unicam aequationem affert

$$0 = 1 + \tfrac{1}{3}\beta'$$

unde $\beta' = -\frac{1}{3}$, sive $U = uu - \frac{1}{3}$.

III. Pro $n = 2$, functio T determinatur adiumento trium aequationum

Ad determinandam U unica aequatio sufficit

$$0 = 1 + \tfrac{1}{3}\beta'$$

unde $\beta' = -\frac{1}{3}$ sive $U = u^3 - \frac{1}{3}u$.

Attamen hunc modum, qui calculos continuo molestiores adducit, hic ulterius non persequemur, sed ad fontem genuinum solutionis generalis progrediemur.

17.

Proposita fractione continua

$$\varphi = \cfrac{v}{w - \cfrac{v'}{w' - \cfrac{v''}{w'' - \cfrac{v'''}{w''' - \text{etc.}}}}}$$

constat, fractiones continuo magis appropinquantes inveniri per algorithmum sequentem. Formentur duae quantitatum series, V, V', V'', V''' etc., W, W', W'', W''' etc. per hasce formulas

$$\begin{aligned}
V &= 0 & W &= 1\\
V' &= v & W' &= w\\
V'' &= w'V' + vV & W'' &= w'W' + vW\\
V''' &= w''V'' + v'V' & W''' &= w''W'' + v'W'\\
V^{IV} &= w'''V''' + v''V'' & W^{IV} &= w'''W''' + v''W''
\end{aligned}$$

etc.

eritque

$$\begin{aligned}
\frac{V'}{W'} &= \frac{v}{w}\\
\frac{V''}{W''} &= \cfrac{v}{w - \cfrac{v'}{w'}}\\
\frac{V'''}{W'''} &= \cfrac{v}{w - \cfrac{v'}{w' - \cfrac{v''}{w''}}}\\
\frac{V^{IV}}{W^{IV}} &= \cfrac{v}{w - \cfrac{v'}{w' - \cfrac{v''}{w'' - \cfrac{v'''}{w'''}}}}
\end{aligned}$$

et sic porro.

Praeterea constat, vel facile ex ipsis aequationibus praecedentibus confirmatur, esse

$$\begin{aligned}
VW' - V'W &= -v\\
V'W'' - V''W' &= +vv'\\
V''W''' - V'''W'' &= -vv'v''\\
V'''W^{IV} - V^{IV}W''' &= +vv'v''v'''
\end{aligned}$$

etc.

Hinc perspicuum est, seriei

$$\frac{V'}{WW'},\quad \frac{V''}{W'W''},\quad \frac{V'''}{W''W'''},\quad \frac{V^{IV}}{W'''W^{IV}}\quad \text{etc.}$$

terminum primum esse $= \frac{v}{ww'}$, summam duorum terminorum primum $= \frac{V''}{WW''}$, summam trium terminorum primum $= \frac{V'''}{WW'''}$, summam quatuor terminorum primum $= \frac{V^{IV}}{WW^{IV}}$, et sic porro; quocirca series ipsa vel in infinitum vel usque dum abrumpatur continuata ipsam fractionem continuam φ exprimet. Simul hinc habetur differentia inter φ atque singulas fractiones appropinquantes $= \frac{V}{W}, \frac{V'}{W'}, \frac{V''}{W''}, \frac{V'''}{W'''}$ etc.

E formula 33 art. 14 Disquisitionum generalium circa seriem infinitam mutando x in $-x$, facile obtinemus transformationem seriei

$$\varphi = \frac{1}{u} + \frac{1}{uu} + \frac{1}{u^3} + \frac{1}{u^4} + \text{etc.}$$

in fractionem continuam sequentem

$$\varphi = \cfrac{1}{u - \cfrac{\frac{1\cdot 2}{1\cdot 3}}{u - \cfrac{\frac{2\cdot 3}{3\cdot 5}}{u - \cfrac{\frac{3\cdot 4}{5\cdot 7}}{u - \cfrac{\frac{4\cdot 5}{7\cdot 9}}{u - \text{etc.}}}}}}$$

ita ut habeatur

$$\begin{array}{lllll} v = 1, & v' = -\frac{1\cdot 2}{1\cdot 3}, & v'' = -\frac{2\cdot 3}{3\cdot 5}, & v''' = -\frac{3\cdot 4}{5\cdot 7}, & v^{IV} = -\frac{4\cdot 5}{7\cdot 9} \quad \text{etc.} \\ w = u, & w' = u, & w'' = u, & w''' = u, & w^{IV} = u \quad \text{etc.} \end{array}$$

Hinc pro V, V', V'', V''' etc., W, W', W'', W''' etc. nanciscimur valores sequentes

$$\begin{array}{rr} V = 0, & W = 1 \\ V' = 1, & W' = u \\ V'' = u, & W'' = uu - \frac{1\cdot 2}{1\cdot 3} \\ V''' = uu - \frac{2}{3}, & W''' = u^3 - \frac{5}{3}u \\ V^{IV} = u^3 - \frac{11}{5}u, & W^{IV} = u^4 - \frac{20}{5}uu + \frac{8}{5}\cdot\frac{2}{3} \quad \text{etc.} \end{array}$$

Tabula I.

Logarithmi briggici functionis Πz, pro z a 0 usque ad 1 per singulas partes centesimas.

z	$\log \Pi z$	z	$\log \Pi z$
0,00	0,0000000000 0000000000	0,50	9,9475449406 8308573196
0,01	9,9975287306 5869172624	0,51	9,9477236538 6182228429
0,02	9,9951278719 8879034144	0,52	9,9479426085 7494550351
0,03	9,9927964208 8883589748	0,53	9,9482014538 7500065798
0,04	9,9905334004 0842900595	0,54	9,9484998446 4251966174
0,05	9,9883378587 9012046216	0,55	9,9488374414 4659973817
0,06	9,9862088685 5581945437	0,56	9,9492139104 0978143536
0,07	9,9841455256 3523567773	0,57	9,9496289230 7706494873
0,08	9,9821469485 3403172902	0,58	9,9500821562 8891076887
0,09	9,9802122775 3951136603	0,59	9,9505732920 5807738191
0,10	9,9783406739 6180754713	0,60	9,9511020174 5015512544
0,11	9,9765313194 0866250820	0,61	9,9516680244 6766136244
0,12	9,9747834150 9201128963	0,62	9,9522710099 3756789859
0,13	9,9730961811 6469083029	0,63	9,9529106754 0213704917
0,14	9,9714688560 8569966779	0,64	9,9535867270 1294797674
0,15	9,9699006960 1252903489	0,65	9,9542988754 2799988466
0,16	9,9683909742 1917527943	0,66	9,9550468357 1178337730
0,17	9,9669389805 3852656982	0,67	9,9558303272 3821579829
0,18	9,9655440208 2789424567	0,68	9,9566490735 9634064632
0,19	9,9642054164 5653136262	0,69	9,9575028024 9869525351
0,20	9,9629225038 1404835193	0,70	9,9583912456 9225480685
0,21	9,9616946338 3869862929	0,71	9,9593141388 7186450668
0,22	9,9605211715 6456577252	0,72	9,9602712215 9607519880
0,23	9,9594014956 8673884734	0,73	9,9612622372 0530119641
0,24	9,9583349981 4361387302	0,74	9,9622869327 4222223320
0,25	9,9573210837 1507540011	0,75	9,9633450588 7435456829
0,26	9,9563591696 3881435774	0,76	9,9644363698 1871920339
0,27	9,9554486852 3498063412	0,77	9,9655606232 6853798084
0,28	9,9545890715 5360828076	0,78	9,9667175803 2189101417
0,29	9,9537797810 2903856417	0,79	9,9679070054 1227146665
0,30	9,9530202771 4980077695	0,80	9,9691286662 4097614416
0,31	9,9523100341 4034352140	0,81	9,9703823337 1127271250
0,32	9,9516485366 5449703876	0,82	9,9716677818 6428658993
0,33	9,9510352794 8014390879	0,83	9,9729847878 1655271065
0,34	9,9504697672 5460261315	0,84	9,9743331316 9917940601
0,35	9,9499515141 9025401627	0,85	9,9757125965 9857361442
0,36	9,9494800438 0996487612	0,86	9,9771229684 9867851092
0,37	9,9490548886 9188515282	0,87	9,9785640362 2467644771
0,38	9,9486755902 2321722697	0,88	9,9800355913 8811182162
0,39	9,9483416983 6257525751	0,89	9,9815374283 3339013630
0,40	9,9480527714 1057187897	0,90	9,9830693440 8561111078
0,41	9,9478083757 8828733374	0,91	9,9846311382 9969520321
0,42	9,9476080858 2329302469	0,92	9,9862226132 1076437381
0,43	9,9474514835 4291742066	0,93	9,9878435735 8573930651
0,44	9,9473381584 7445730981	0,94	9,9894938266 7611664682
0,45	9,9472677074 5205163055	0,95	9,9911731821 7189109803
0,46	9,9472397344 2994856529	0,96	9,9928814521 5658844947
0,47	9,9472538503 0190930853	0,97	9,9946184510 6337679375
0,48	9,9473096727 2650396072	0,98	9,9963839956 3222432515
0,49	9,9474068259 5806639475	0,99	9,9981779048 6807320161

Tabula II.

Valores functionis Ψz, pro z a 0 usque ad 1 per singulas partes centesimas.

z	Ψz	z	Ψz
0,00	−0,5772156649 0153286061	0,50	+0,0364899739 7857652060
0,01	−0,5608854578 68674498	0,51	+0,0550221145 66077154
0,02	−0,5447893104 56179789	0,52	+0,0641673073 40635189
0,03	−0,5289210872 85430502	0,53	+0,0732336936 51781439
0,04	−0,5132748789 16830312	0,54	+0,0822225675 44444623
0,05	−0,4978449912 99870371	0,55	+0,0911351925 79551622
0,06	−0,4826259358 14825705	0,56	+0,0999728024 35011042
0,07	−0,4676124198 67553632	0,57	+0,1087366022 73476253
0,08	−0,4527993380 01712885	0,58	+0,1174277690 68980212
0,09	−0,4381817634 95334764	0,59	+0,1260474527 58445210
0,10	−0,4237549404 11076796	0,60	+0,1345967771 19958383
0,11	−0,4095142760 71694248	0,61	+0,1430768403 02762795
0,12	−0,3954553339 34292807	0,62	+0,1514887158 46861164
0,13	−0,3815738268 38792064	0,63	+0,1598334528 23688336
0,14	−0,3678656106 07749546	0,64	+0,1681120775 35096466
0,15	−0,3543266779 76279272	0,65	+0,1763255932 71894293
0,16	−0,3409531528 32261794	0,66	+0,1844749812 14883540
0,17	−0,3277412847 48392299	0,67	+0,1925612014 14168016
0,18	−0,3146874437 88860621	0,68	+0,2005851930 88779580
0,19	−0,3017881155 74610030	0,69	+0,2085478748 93861542
0,20	−0,2890398965 92188296	0,70	+0,2164501461 64887544
0,21	−0,2764394897 32192051	0,71	+0,2242928871 96501033
0,22	−0,2639837000 44220200	0,72	+0,2320769593 31568639
0,23	−0,2516694306 96100107	0,73	+0,2398032061 83820641
0,24	−0,2394936791 25936794	0,74	+0,2474724535 94485409
0,25	−0,2274535333 76265408	0,75	+0,2550855103 45479708
0,26	−0,2155461686 00265182	0,76	+0,2626431686 49528903
0,27	−0,2037688477 30623157	0,77	+0,270146043 57135499
0,28	−0,1921188983 02221732	0,78	+0,2775953776 64059720
0,29	−0,1805937494 20369178	0,79	+0,2849914332 06401469
0,30	−0,1691908888 66799656	0,80	+0,2923351011 48881123
0,31	−0,1579078803 36141874	0,81	+0,2996270965 23498793
0,32	−0,1467423567 95996017	0,82	+0,3068681204 46740214
0,33	−0,1356920179 64169332	0,83	+0,3140588602 42043086
0,34	−0,1247546278 97003946	0,84	+0,3211999895 13771306
0,35	−0,1139280126 83088296	0,85	+0,3282921690 49890838
0,36	−0,1032100582 36977615	0,86	+0,3353360466 98467139
0,37	−0,0925987081 87861259	0,87	+0,3423322577 45414956
0,38	−0,0820919618 58406487	0,88	+0,3492814254
0,39	−0,0716878723 29281510	0,89	+0,3561841611
0,40	−0,0613845445 85116146	0,90	+0,3630410646
0,41	−0,0511801337 37897756	0,91	+0,3698527244
0,42	−0,0410728433 24024375	0,92	+0,3766197179
0,43	−0,0310609236 71447052	0,93	+0,3833426119
0,44	−0,0211426703 33530475	0,94	+0,3900219627
0,45	−0,0113164225 86445845	0,95	+0,3966583163
0,46	−0,0015805619 87083418	0,96	+0,4032522088
0,47	+0,0080664890 11364893	0,97	+0,4098041664
0,48	+0,0176262683 88849468	0,98	+0,4163147060
0,49	+0,0271002758 35486201	0,99	+0,4227843350
1,00	+0,4227843350 98467139		